

A FAMILY OF INTEGRALS FOR π WITH THE SYMMETRY $z \mapsto 2/z$, AND THE BOUND $\mu(\pi) \leq 6.0446$

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ABSTRACT. We show that the irrationality exponent of π is at most 6.0446: for every $\mu > 6.0446$ the inequality $|\pi - a/b| > b^{-\mu}$ holds for all integers a and all sufficiently large integers b . The best published bound is 7.103205... (Zeilberger and Zudilin, 2020). The proof uses the integrals

$$J_n = -i \int_{1-i}^{1+i} \left(\frac{((z-1)(z-2))^5 (z^2 - 2z + 2)^3}{z^4 (z^2 - 2)^4} \right)^n \frac{dz}{z} = u_n + v_n \pi \quad (n = 1, 2, \dots),$$

whose integrand is invariant under $z \mapsto 2/z$. We prove that u_n and v_n are rational numbers whose denominators divide $D_n = \text{lcm}(1, \dots, 4n)/\Phi_n$, where Φ_n is an explicit product of “removable” primes; that $v_n = e^{r n + o(n)}$ with $r \leq 10.540357$; and that $|J_n| \leq 4e^{-6.530134 n}$. The first integral is $J_1 = \frac{7345}{2}(\pi - \frac{355}{113})$. The paper is written for readers who are not specialists in Diophantine approximation: the steps are derived in detail (one of them, a polynomial inequality in one real variable, is verified by computer), $n = 1$ is used as a running example, and Appendix A contains the computations line by line.

Status. To our knowledge this is a new result. It was generated by AI systems working under human direction, and it has *not yet been refereed by a human expert*. The complete proof of the bound has been formalised in Lean 4 with Mathlib: the machine-checked theorem states the bound 6.0446 under a single hypothesis, the prime number theorem in the form $\theta(x) \sim x$, which is not yet available in Mathlib (Section 10).

1. INTRODUCTION

1.1. How well can π be approximated by fractions? Every real number can be approximated by rational numbers a/b ; the interesting question is how small the error can be in terms of the denominator b . For every irrational number θ there are infinitely many fractions with $|\theta - a/b| < b^{-2}$ (Dirichlet). For π , the fraction 355/113 is even better than that: $|\pi - 355/113| = 2.667 \dots \cdot 10^{-7} \approx 113^{-3.20}$. Does this happen infinitely often, with exponents like 3.2 or larger? Nobody knows. The quantity that measures it is the following.

Definition 1.1. Let θ be a real irrational number. The *irrationality exponent* $\mu(\theta)$ is the infimum of all real numbers μ with the following property: there is b_0 such that

$$\left| \theta - \frac{a}{b} \right| > \frac{1}{b^\mu} \quad \text{for all integers } a \text{ and all integers } b \geq b_0.$$

(If no such μ exists, $\mu(\theta) = \infty$.)

By Dirichlet’s theorem $\mu(\theta) \geq 2$ for every irrational θ . Equality holds for almost all real numbers (in the sense of Lebesgue measure), for all irrational algebraic numbers (Roth’s theorem) and for e . It is generally expected that $\mu(\pi) = 2$ as well, but what can be proved is much weaker. An inequality $\mu(\pi) \leq \mu_0$ is a quantitative form of the irrationality of π : it says that $|\pi - a/b| > b^{-\mu_0 - \delta}$ for every $\delta > 0$ and all large b . Table 1 lists the bounds obtained since Mahler’s first result.

year	author	$\mu(\pi) \leq$	method
1953	Mahler	42	Hermite's approximations to e^z
1974	Mignotte	20	refinement of Mahler's method
1982	G. Chudnovsky	19.8899944...	Hermite–Padé approximations
1993	Hata [3]	13.39...	modified contour integral
1993	Hata [4]	8.0161	the integrals $\int ((z-1)(z-2)(z-1-i))^{2n} z^{-3n-1} dz$
2008	Salikhov [8, 9]	7.6063	symmetrised complex integral
2020	Zeilberger–Zudilin [10]	7.1032	another member of Salikhov's family
2026	Bai [2] (preprint)	7.1019	deformation inside the same family
2026	this paper	6.0446	integrals with the symmetry $z \mapsto 2/z$

TABLE 1. Upper bounds for the irrationality exponent of π . The first four entries and their attributions are taken from the introduction of [4]. For the fourth entry that introduction gives the value 13.394 (citing [3] as “to appear”), whereas the published paper [3] is reported to state 13.398; we have not seen it, and the table shows the common digits. Mahler proved $|\pi - a/b| > b^{-42}$ for all $b \geq 2$ and remarked that 42 can be replaced by 30 for large b . A preprint [6] conjectures that 7.1019 is the limit of the family of [9, 10, 2].

1.2. Small linear forms. All entries of Table 1 are proved in the same way. One constructs integers q_n and p_n ($n = 1, 2, \dots$) such that the *linear form* $q_n\pi - p_n$ is very small while q_n is not too large. The following lemma, in the form given by Hata, converts the two rates into a bound for the exponent.

Lemma 1.2 (Hata [4, Remark 2.1]). *Let θ be a real irrational number and let q_n, p_n ($n \geq 1$) be integers such that*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log |q_n| = \sigma \quad \text{and} \quad \limsup_{n \rightarrow \infty} \frac{1}{n} \log |q_n\theta - p_n| \leq -\tau$$

for some real numbers $\sigma > 0$ and $\tau > 0$. Then $\mu(\theta) \leq 1 + \sigma/\tau$.

The idea is simple. Suppose that a/b is very close to θ . For every n ,

$$q_n a - p_n b = b(q_n\theta - p_n) - q_n(b\theta - a)$$

is an integer. Take n just large enough that $b|q_n\theta - p_n| < \frac{1}{2}$, that is $n \approx (\log b)/\tau$; then $|q_n| \approx b^{\sigma/\tau}$. If the integer on the left is not zero, its absolute value is at least 1, so $|q_n| |b\theta - a| > \frac{1}{2}$ and therefore $|\theta - a/b| > 1/(2b|q_n|) \approx b^{-1-\sigma/\tau}$. The complete proof (which also handles the case where the integer vanishes) is given in Appendix A.9. Note that only an *upper* bound for the small quantity $|q_n\theta - p_n|$ is required.

So the game is to find linear forms with a small ratio σ/τ . For the record of Zeilberger and Zudilin, $\sigma = 11.6138\dots$ and $\tau = 1.9029\dots$; in this paper $\sigma \leq 14.2464$ and $\tau \geq 2.82409$.

1.3. The integrals of this paper. Put

$$(1.1) \quad N(z) = ((z-1)(z-2))^5 (z^2 - 2z + 2)^3, \quad f(z) = \frac{N(z)}{z^4 (z^2 - 2)^4},$$

and, for every integer $n \geq 1$,

$$(1.2) \quad J_n = -i \int_{1-i}^{1+i} f(z)^n \frac{dz}{z},$$

where the integral is taken along the straight segment from $1 - i$ to $1 + i$. The polynomial N vanishes at 1, 2 and $1 \pm i$; the function f has poles at 0 and $\pm\sqrt{2}$; and $f(2/z) = f(z)$ (Lemma 3.1). The factor $-i$ makes J_n real.

To state the arithmetic result we need a definition. For integers $m \geq 0$ and $p \geq 1$ we write $m \bmod p \in \{0, 1, \dots, p-1\}$ for the remainder of m on division by p ; $[x]$ and $\{x\} = x - [x]$ are the integer part and the fractional part of a real number x .

Definition 1.3. Let $n \geq 1$. A prime number p is *removable for n* if

$$p^2 > 4n, \quad [4n/p] \text{ is even}, \quad (5n \bmod p) + (3n \bmod p) < (4n \bmod p).$$

We denote by Φ_n the product of all removable primes $p \leq 4n$, and we put

$$D_n = \frac{\text{lcm}(1, 2, \dots, 4n)}{\Phi_n}.$$

Since a removable prime $p \leq 4n$ satisfies $p^2 > 4n$, it divides $\text{lcm}(1, \dots, 4n)$ exactly once; so D_n is an integer. We shall see (Lemma 8.1) that a prime $p > \sqrt{4n}$ is removable if and only if the fractional part of n/p lies in $[\frac{1}{5}, \frac{1}{4}) \cup [\frac{2}{3}, \frac{3}{4})$. For example $\Phi_1 = 1$, $\Phi_2 = 3$ (because $\{2/3\} = \frac{2}{3}$) and $\Phi_{21} = 17 \cdot 29 \cdot 31$.

Theorem 1.4 (the linear forms). *For every integer $n \geq 1$:*

- (a) $J_n = u_n + v_n\pi$ with rational numbers u_n and v_n . More precisely $v_n = \frac{1}{2}c_n$, where c_n is the coefficient of z^{4n} in the power series expansion of $N(z)^n(z^2 - 2)^{-4n}$ at $z = 0$, and c_n is a positive integer.
- (b) $D_n u_n$ and $D_n v_n$ are integers.
- (c) The limit $r = \lim_{n \rightarrow \infty} \frac{1}{n} \log v_n$ exists, and $8.9 < r \leq 10.540357$.
- (d) $|J_n| \leq 4\rho^n$ with $\rho = 0.001458809 < e^{-6.530134}$.
- (e) $\lim_{n \rightarrow \infty} \frac{1}{n} \log D_n = 4 - \varpi$, where

$$\varpi = \sum_{k=0}^{\infty} \frac{1}{(3k+2)(4k+3)} + \sum_{k=1}^{\infty} \frac{1}{(4k+1)(5k+1)} = 0.2939598992 \dots$$

Theorem 1.5. *The irrationality exponent of π satisfies $\mu(\pi) \leq 6.0446$.*

Theorem 1.5 follows from Theorem 1.4 and Lemma 1.2 applied to the integers $q_n = D_n v_n$ and $p_n = -D_n u_n$, for which $q_n \pi - p_n = D_n J_n$: one gets $\sigma = r + (4 - \varpi) \leq 14.2464$ and $\tau \geq 6.530134 - (4 - \varpi) \geq 2.82409$, and $1 + 14.2464/2.82409 < 6.0446$. The prime number theorem is used only to determine the size of the multiplier (part (e), and Proposition 8.3 below). The details are in Section 9; there we work with a slightly larger multiplier $D_n^{(K)}$ (a multiple of D_n in which only the removable primes of finitely many “windows” are removed), for which part (e) is needed only in a truncated form. This is the version that has been formalised, and it gives the same bound. The exact values of the three rates are discussed in Remark 9.2; they would give $6.04458594 \dots$, so nothing is lost in the rounding.

Example 1.6 ($n = 1$). For $n = 1$ one finds (Example 4.8)

$$J_1 = \frac{7345\pi - 23075}{2} = \frac{7345}{2} \left(\pi - \frac{355}{113} \right) = -0.000979691 \dots,$$

because $7345 = 65 \cdot 113$ and $23075 = 65 \cdot 355$. So the first integral reproduces the approximation $355/113$. Here $D_1 = \text{lcm}(1, 2, 3, 4) = 12$, and indeed $12u_1 = -138450$ and $12v_1 = 44070$ are integers. Table 2 shows the first six forms.

n	u_n	v_n	D_n	Φ_n
1	$-\frac{23075}{2}$	$\frac{7345}{2}$	12	1
2	$-\frac{4338189943}{14}$	$\frac{197269821}{2}$	280	3
3	$-\frac{11065298371140502}{1155}$	$\frac{6099036995857}{2}$	27720	1
4	$-\frac{14141220324520524847492}{45045}$	$\frac{199857486158198941}{2}$	720720	1
5	$-\frac{173152818047406076172911685}{16302}$	3380950423572104245485	33256080	7
6	$-\frac{743759148883149273178854708603850}{2028117}$	$\frac{233463740039798755103513775}{2}$	1070845776	5

TABLE 2. The linear forms $J_n = u_n + v_n\pi$ for $n \leq 6$, with the multiplier $D_n = \text{lcm}(1, \dots, 4n)/\Phi_n$. In each case the denominator of u_n divides D_n ; for instance $1155 = 3 \cdot 5 \cdot 7 \cdot 11$ divides 27720, and $16302 = 2 \cdot 3 \cdot 11 \cdot 13 \cdot 19$ divides $33256080 = \text{lcm}(1, \dots, 20)/7$.

1.4. What is new. The integral (1.2) is a member of a three-parameter family, in which the factors $(z-1)(z-2)$, z^2-2z+2 and z^2-2 carry exponents e_1n, e_2n, dn ; here $(e_1, e_2, d) = (5, 3, 4)$ (Appendix B). The member $(3, 3, 2)$ gives, up to rational factors, the same linear forms as the integral of Zeilberger and Zudilin (this was tested numerically, Appendix C). The improvement comes from giving the two pairs of zeros $\{1, 2\}$ and $\{1 \pm i\}$ *different* exponents. Hata’s integrand of 1993 has the zeros 1, 2 and $1+i$ and a pole at 0 only; the integrand used here has in addition the zero $1-i$ and the two poles $\pm\sqrt{2}$, which are fixed by the symmetry $z \mapsto 2/z$ and contribute no logarithm to the integral (Remark 4.4). Three facts make the construction work: the only “bad” prime is 2, and its contribution to the denominators is only $2^{O(\log n)}$ (Section 5.4); many primes up to $4n$ can be removed from the denominators, their product being $e^{0.2939\dots n}$ (Sections 5.3, 8); and the integral is exponentially smaller than the straight path of integration suggests (Section 7). A web search found no earlier occurrence of these integrals, but the search of the literature was limited.

1.5. Status of the result. The reader should be aware of the following.

- To our knowledge the result is new; it would improve a record that has stood since 2020 by a wide margin. It has *not* been refereed by a human expert, and the search of the literature was limited.
- The work is AI-generated: the integrals were found in a computer search directed by a human and carried out by AI systems (large language models with access to exact computer algebra), and the same systems wrote the proofs, two independent audits, the formalisation and this text.
- The proof has been formalised in Lean 4 with Mathlib. The formal statement is Theorem 1.5 under the hypothesis $\theta(x) \sim x$ (the prime number theorem, which is not yet part of Mathlib); every other step of the proof of Theorem 1.5, including all numerical estimates that it uses, is machine-checked. Statements of this paper that are not needed for Theorem 1.5 are *not* formalised: Theorem 1.4(e), the lower bound $8.9 < r$ in Theorem 1.4(c), the exact values of the constants, and Appendices B–D. Section 10 describes exactly what has been verified. The proofs in this paper follow the formalised route; where it differs from the classical one we say so in a remark.

1.6. Organisation. Section 2 gives the proof in one page and a table of notation. Section 3 describes the integrand. Sections 4–8 prove the five parts of Theorem 1.4, and Section 9 proves Theorem 1.5. Section 10 describes the formal verification and Section 11 some numerical checks.

The main text gives every statement, the idea of its proof and a worked example; whenever it says that something is a direct computation, the computation is written out in Appendix A. Material for specialists (other members of the family, the relation to the integrals of Zeilberger and Zudilin, companion forms in $1, \pi, \log 2$) is collected in Appendices B–D.

1.7. Accompanying material. This paper is distributed together with a folder of accompanying material, to which we refer in several places. It contains: the Lean project (directory `lean/`; Section 10); the programs that recompute the numbers printed in this paper (directory `scripts/`; Section 11); the *original draft* of the proof (file `K2H_PROOF.md`), which treats the whole family of Appendix B by the classical route and from which Appendices B–D are taken; two *audits* of that draft by AI systems (files `AUDIT_PROOF.md` and `AUDIT_NUMERIC.md`); and a *companion note* on linear forms in $1, \pi, \log 2$ (file `K2H_LOG2_NOTE.md`). At the time of writing this material is kept in a private repository and has no public address. The proof of Theorem 1.5 given in Sections 3–9 and Appendix A does not depend on these documents, with one exception: the 99 rational numbers of the certificate in Proposition 7.1 are too long to print. They are contained in the file `lean/MuPi/K2H/ContourCert.lean`, and Appendix A.7 describes how to recompute them.

2. THE PROOF IN ONE PAGE

Write $h_n(z) = f(z)^n/z$ for the integrand of (1.2) and $\Omega = \mathbb{C} \setminus \{0, \sqrt{2}, -\sqrt{2}\}$ for the set where it is defined.

- 1. Primitive pairs.** A pair (G, c) , with G a holomorphic function on Ω and c a complex number, is called a *primitive pair* if $G'(z) = h_n(z) - c/z$ on Ω . Any two primitive pairs have the same c and the same difference $G(1+i) - G(1-i)$, and the fundamental theorem of calculus gives

$$J_n = -i(G(1+i) - G(1-i)) + \frac{c\pi}{2},$$

because $\log z$ increases by $i\pi/2$ along the segment. So $u_n = -i(G(1+i) - G(1-i))$ and $v_n = c/2$ can be computed from *any* primitive pair (Section 4.1). The whole proof consists in exhibiting several primitive pairs, each adapted to one purpose.

- 2. Rationality.** In the variables $W = z + 2/z$ and $Y = z - 2/z$ one has $f = (W - 3)^5(W - 2)^3/Y^4$, and a two-term recursion integrates every function $W^j Y^{-2m}/z$. This gives a primitive pair with rational data, hence $u_n, v_n \in \mathbb{Q}$ (Section 4.2).
- 3. Denominators.** For every prime p we exhibit a primitive pair that is “ p -integral” up to an explicit power of p : for odd p , the partial fraction decomposition of h_n , integrated term by term, has denominators $\leq 4n$ (Section 5.2); for a removable prime, an identity $h_n = (g^p G_{\text{red}})' - p g^{p-1} g' G_{\text{red}}$ with suitable rational functions g and G_{red} shows that the factor p in those denominators cancels (Section 5.3); for $p = 2$, a Möbius change of variable $z \mapsto w$ turns h_n into a combination of elementary integrands whose primitives are divisible by explicit powers of 2 (Section 5.4). Together: $D_n u_n, D_n v_n \in \mathbb{Z}$ (Section 5.5).
- 4. Growth.** After the substitution $z \mapsto -z$ the number c_n is the coefficient of z^{4n} in the n -th power of a power series with non-negative coefficients. Hence $c_{m+n} \geq c_m c_n$, the limit r exists (Fekete’s lemma), and $c_n \leq f(-x)^n$ for $0 < x < \sqrt{2}$; with $x = 0.3913$ this gives $r \leq 10.540357$ (Section 6).
- 5. Decay.** The primitive $G + c \text{Log } z$ (with the principal logarithm Log) also computes the integral of h_n along the polygon $1 - i \rightarrow \bar{z}_s \rightarrow 1 \rightarrow z_s \rightarrow 1 + i$ through a point z_s next to a saddle point of $|f|$. On this polygon $|f| \leq \rho$, which is a polynomial inequality in one real variable and is certified by a Bernstein expansion. Hence $|J_n| \leq 4\rho^n$ (Section 7).

6. The multiplier. By the prime number theorem $\log \text{lcm}(1, \dots, 4n) \sim 4n$, and the removable primes are those in the “windows” $\{n/p\} \in [\frac{1}{5}, \frac{1}{4}) \cup [\frac{2}{3}, \frac{3}{4})$; so $\frac{1}{n} \log D_n \rightarrow 4 - \varpi$ (Section 8).

7. Conclusion. Hata’s lemma with $q_n = D_n v_n$, $p_n = -D_n u_n$; in the formalised version D_n is replaced by a multiple $D_n^{(K)}$ that is easier to estimate (Section 9).

The numbers fit together as shown in Table 3; Table 4 lists the notation.

rate	meaning	bound used	exact value
r	growth of v_n	$r \leq 10.540357$	$10.5403551 \dots$
s	decay of J_n : $ J_n \leq 4e^{-sn}$	$s = 6.530134$	$6.5301361 \dots$ (saddle point)
C	size of the multiplier: $e^{Cn+o(n)}$	$C \leq 3.706043$	$4 - \varpi = 3.7060401 \dots$
$\sigma = r + C$	growth of q_n	$\sigma \leq 14.2464$	$14.2463952 \dots$
$\tau = s - C$	decay of $q_n \pi - p_n$	$\tau \geq 2.82409$	$2.8240960 \dots$
$1 + \sigma/\tau$	bound for $\mu(\pi)$	< 6.0446	$6.0445859 \dots$

TABLE 3. The three rates and how they combine. The column “bound used” contains what is proved, and formally verified, in this paper; there the multiplier is $D_n^{(K)}$ with $K = 10^5$ (Section 8), for which $C = 4 - \varpi_K \leq 3.706043$. The last column contains the values discussed in Remark 9.2; they belong to the multiplier D_n and are not needed for the proof (those of r and s are not proved in this paper).

symbol	meaning
$\mu(\theta)$	irrationality exponent (Definition 1.1)
$N, f, h_n = f^n/z$	numerator, integrand (1.1), and the function that is integrated
Ω	$\mathbb{C} \setminus \{0, \sqrt{2}, -\sqrt{2}\}$
$J_n = u_n + v_n\pi$	the integral (1.2) and its rational coefficients
$c_n = 2v_n$	coefficient of z^{4n} in $N(z)^n(z^2 - 2)^{-4n}$; the “residue of h_n at 0”
(G, c)	primitive pair: $G' = h_n - c/z$ on Ω (Definition 4.1)
W, Y	$W = z + 2/z, Y = z - 2/z; f = (W - 3)^5(W - 2)^3/Y^4$
$X_j^{(m)}$	$W^j Y^{-2m}/z$ (Section 4.2)
$[x], \{x\}, m \bmod p$	integer part, fractional part, remainder in $\{0, \dots, p-1\}$
Φ_n, D_n	product of the removable primes $\leq 4n$; $D_n = \text{lcm}(1, \dots, 4n)/\Phi_n$
$\Phi_n^{(K)}, D_n^{(K)}$	product of the primes in the windows (8.1) with $k < K$; $D_n^{(K)} = \text{lcm}(1, \dots, 4n)/\Phi_n^{(K)}$; $K = 10^5$ in Section 9
$R_p, \mathbb{Z}_{(p)}$	p -integers of $\mathbb{C}$ (Definition 5.1); $\mathbb{Z}_{(p)} = R_p \cap \mathbb{Q}$
$\mathcal{A}(R)$	rational functions $M(z)/(z^{e_0}(z - \sqrt{2})^{e_+}(z + \sqrt{2})^{e_-})$ with $M \in R[z]$
$\varepsilon, \varkappa$	$\varepsilon = \sqrt{2} - 1, \varkappa = 1 + \varepsilon^2 = 2\sqrt{2}\varepsilon$
w	the coordinate $w = i\varepsilon(\sqrt{2} + z)/(\sqrt{2} - z)$ (Section 5.4)
$\mathfrak{L}_k$	$(\sqrt{2})^k R_2$: numbers “divisible by $(\sqrt{2})^k$ ” (Definition 5.16)
$s_2(a), L$	sum of the binary digits of a ; $L = \lfloor \log_2(4n) \rfloor$
r, s, C	the three rates of Table 3
x_0, λ_3	minimum point and minimal value of $f(-x)$ on $0 < x < \sqrt{2}$ (Remark 6.4)
ρ, z_s, Γ	$\rho = 0.001458809$; $z_s = \frac{19787}{16384} + \frac{32371}{65536}i$; the polygon $1 - i \rightarrow \bar{z}_s \rightarrow 1 \rightarrow z_s \rightarrow 1 + i$
z_1	saddle point of $ f $ next to z_s (Remark 7.4)
$\theta(x), \psi(x)$	Chebyshev’s functions (Section 8)
ϖ, ϖ_K	the constant of Theorem 1.4(e) and its partial sum (8.2)
σ, τ	the two rates in Hata’s lemma

TABLE 4. Notation.

3. THE INTEGRAND

3.1. Zeros, poles and the symmetry. The numerator $N(z) = (z-1)^5(z-2)^5(z-1-i)^3(z-1+i)^3$ has degree 16, so $f = N/(z^4(z-\sqrt{2})^4(z+\sqrt{2})^4)$ has zeros of order 5 at 1 and 2, zeros of order 3 at the end points $1 \pm i$ of the path, and poles of order 4 at $0, \sqrt{2}, -\sqrt{2}$ (and at ∞). Figure 1 shows these points.

Lemma 3.1 (symmetry). $f(2/z) = f(z)$ for all $z \in \Omega$.

Proof. Replace z by $2/z$ in each factor:

$$\left(\frac{2}{z}-1\right)\left(\frac{2}{z}-2\right) = \frac{2(z-1)(z-2)}{z^2}, \quad \left(\frac{2}{z}\right)^2 - 2 \cdot \frac{2}{z} + 2 = \frac{2(z^2 - 2z + 2)}{z^2}, \quad \left(\frac{2}{z}\right)^2 - 2 = -\frac{2(z^2 - 2)}{z^2}.$$

Hence the numerator of $f(2/z)$ is $2^{5+3}N(z)/z^{16}$ and the denominator is $(2/z)^4 \cdot 2^4(z^2 - 2)^4/z^8 = 2^8 z^{-12}(z^2 - 2)^4$; the quotient is $f(z)$. $\square$

The map $z \mapsto 2/z$ exchanges 1 and 2, exchanges $1+i$ and $1-i$, exchanges 0 and ∞ , and fixes $\pm\sqrt{2}$. It maps the circle $|z| = \sqrt{2}$, which passes through $1 \pm i$ and $\pm\sqrt{2}$, to itself. The integrand

of (1.2) is built to respect this symmetry; this is the reason why the two fixed points $\pm\sqrt{2}$ will turn out to be harmless poles (they contribute no logarithm, Remark 4.4).

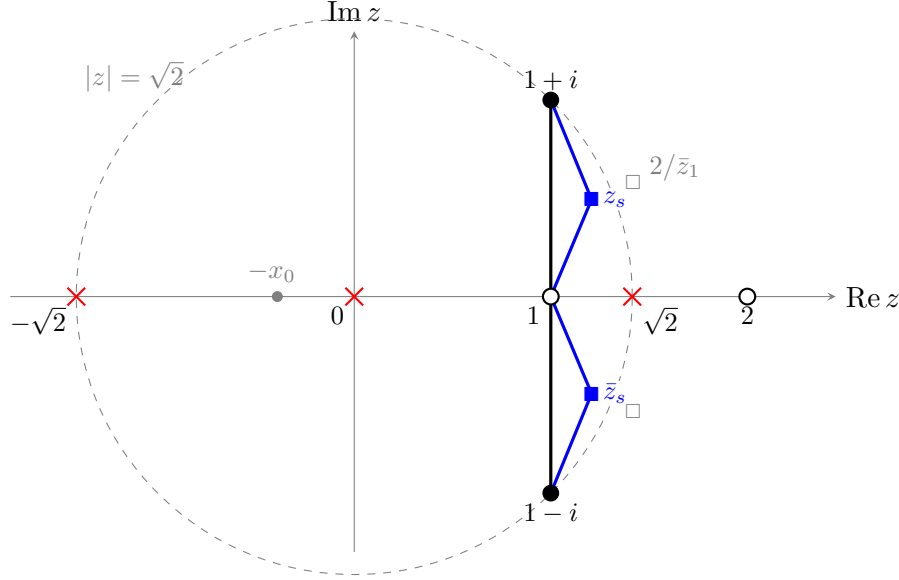


FIGURE 1. The z -plane. Crosses: the poles $0, \pm\sqrt{2}$ of f . Open circles: the zeros $1, 2$ (order 5). Black dots: the zeros $1 \pm i$ (order 3), which are the end points of the path. Black segment: the path of integration in (1.2). Blue: the polygon $1 - i \rightarrow \bar{z}_s \rightarrow 1 \rightarrow z_s \rightarrow 1 + i$ of Section 7; its vertex z_s is a rational point at distance less than 10^{-5} from the saddle point $z_1 = 1.2077099\dots + 0.4939472\dots i$ of $|f|$. The symmetry $z \mapsto 2/z$ maps the dashed circle to itself and $z_1, \bar{z}_1$ to the saddle points $2/z_1, 2/\bar{z}_1$, which are marked by grey squares. The grey dot is the point $-x_0 = -0.39126\dots$ where $|f|$ is minimal on the interval $(-\sqrt{2}, 0)$ (Section 6).

3.2. The invariant coordinates W and Y . The symmetry suggests working with a function that is invariant under $z \mapsto 2/z$ and one that changes sign:

$$(3.1) \quad W = z + \frac{2}{z}, \quad Y = z - \frac{2}{z}.$$

Lemma 3.2. *On $\mathbb{C} \setminus \{0\}$:*

- (a) $W^2 - Y^2 = 8$;
- (b) $z \frac{dW}{dz} = Y$ and $z \frac{dY}{dz} = W$;
- (c) $W - 3 = \frac{(z-1)(z-2)}{z}$, $W - 2 = \frac{z^2 - 2z + 2}{z}$, $Y = \frac{z^2 - 2}{z}$, and therefore

$$(3.2) \quad f = \frac{(W-3)^5 (W-2)^3}{Y^4} \quad \text{on } \Omega;$$

- (d) at the end points, $W(1 \pm i) = 2$ and $Y(1 \pm i) = \pm 2i$.

The proof is a direct computation (Appendix A.1). Formula (3.2) explains the shape of f : in the variable W the zeros are at $W = 3$ (that is $z = 1, 2$) and $W = 2$ (that is $z = 1 \pm i$), and $Y^2 = W^2 - 8$ vanishes at $W = \pm 2\sqrt{2}$ (that is $z = \pm\sqrt{2}$).

3.3. The integral is real. Parametrise the segment by $z = 1 + it$, $-1 \leq t \leq 1$, so that $dz = i dt$ and $J_n = \int_{-1}^1 h_n(1 + it) dt$. Since f has real coefficients, $h_n(\bar{z}) = \overline{h_n(z)}$; so the values of the integrand at t and $-t$ are complex conjugates and J_n is real. Numerically $J_1 = -9.7969 \cdot 10^{-4}$, $J_2 = 1.0455 \cdot 10^{-6}$, $J_3 = -1.2241 \cdot 10^{-9}$.

4. THE LINEAR FORM $J_n = u_n + v_n\pi$

4.1. Primitive pairs. If h_n were the derivative of a rational function, J_n would be a difference of two values of that function, and no π would appear. It is not: its partial fraction decomposition contains a term c/z , whose integral is a logarithm, and this logarithm is where π comes from. The following definition isolates this term.

Definition 4.1. Let $n \geq 1$. A *primitive pair* (for n) is a pair (G, c) consisting of a function $G: \Omega \rightarrow \mathbb{C}$ and a number $c \in \mathbb{C}$ such that G is complex differentiable on Ω and

$$G'(z) = \frac{f(z)^n}{z} - \frac{c}{z} \quad \text{for all } z \in \Omega.$$

Lemma 4.2 (uniqueness). *If (G_1, c_1) and (G_2, c_2) are primitive pairs for the same n , then*

$$c_1 = c_2 \quad \text{and} \quad G_1(1+i) - G_1(1-i) = G_2(1+i) - G_2(1-i).$$

Proof. The function $\Delta = G_1 - G_2$ satisfies $\Delta'(z) = (c_2 - c_1)/z$ on Ω . Integrate Δ' over the circle $|z| = \frac{1}{2}$, which lies in Ω : the integral of a derivative over a closed curve is 0, while the integral of $1/z$ is $2\pi i$. Hence $c_1 = c_2$, and then $\Delta' = 0$ on Ω . The strip $\{z : \frac{1}{2} < \operatorname{Re} z < \frac{5}{4}\}$ is convex, is contained in Ω (because $\sqrt{2} > \frac{5}{4}$) and contains both $1+i$ and $1-i$; a function with vanishing derivative on a convex open set is constant, so $\Delta(1+i) = \Delta(1-i)$. Details are in Appendix A.2. $\square$

Lemma 4.3 (the fundamental theorem of calculus). *If (G, c) is a primitive pair for n , then*

$$J_n = -i(G(1+i) - G(1-i)) + c \cdot \frac{\pi}{2}.$$

Proof. Let Log be the principal branch of the logarithm, which is holomorphic on the half-plane $\operatorname{Re} z > 0$ with derivative $1/z$. On the set $\{\operatorname{Re} z > 0\} \setminus \{\sqrt{2}\}$ the function $G + c \operatorname{Log}$ is a primitive of h_n , and the segment from $1-i$ to $1+i$ lies in this set. Therefore

$$\int_{1-i}^{1+i} h_n(z) dz = G(1+i) - G(1-i) + c(\operatorname{Log}(1+i) - \operatorname{Log}(1-i)),$$

and $\operatorname{Log}(1 \pm i) = \frac{1}{2} \log 2 \pm i\frac{\pi}{4}$, so the last bracket equals $i\pi/2$. Multiply by $-i$. $\square$

Consequently, as soon as one primitive pair exists, the numbers

$$(4.1) \quad u_n := -i(G(1+i) - G(1-i)), \quad c_n := c, \quad v_n := \frac{1}{2}c_n$$

do not depend on the pair and satisfy $J_n = u_n + v_n\pi$. This is how u_n and v_n are *defined* in this paper. We shall construct four kinds of primitive pairs: one with rational data (Section 4.2), one for each odd prime (Sections 5.2, 5.3), and one for the prime 2 (Section 5.4). By Lemma 4.2 each of them gives information about the same numbers u_n and c_n .

Remark 4.4 (the classical formulation). Background for readers who have not met residues: a rational function φ can be written as a polynomial plus a sum of terms $q_{a,k}(z-a)^{-k}$ over its poles a (partial fractions); the coefficient $q_{a,1}$ is the *residue* of φ at a . Every term except the residue terms has a rational primitive, and $q_{a,1}/(z-a)$ integrates to $q_{a,1} \log(z-a)$. In this language a primitive pair exists if and only if the residues of h_n at $\pm\sqrt{2}$ vanish, and then c is the residue at 0. Classically one proves the vanishing from the symmetry: the form $h_n dz$ changes

sign under $z \mapsto 2/z$, and $\pm\sqrt{2}$ are fixed points, so the residue there equals its own negative. In the route followed here the vanishing is a by-product of the explicit construction in Section 4.2.

There is a second point. In the original draft (Section 1.7) the argument for the prime 2 compared two representations $A + B\pi = A' + B'\pi$ with A, B, A', B' in $\mathbb{Q}(\sqrt{2})$ and concluded $A = A'$, $B = B'$ “since $\pi \notin \mathbb{Q}(\sqrt{2})$ ”, which uses that π is not a quadratic irrationality (a consequence of its transcendence). Lemma 4.2 replaces this step by an elementary argument; no arithmetic property of π is used anywhere in the proof of Theorem 1.4, and the irrationality of π is used only in Hata’s lemma.

4.2. A primitive pair with rational data. By (3.2),

$$(4.2) \quad h_n = \frac{P_n(W)}{Y^{4n}} \cdot \frac{1}{z}, \quad P_n(W) = ((W-3)^5(W-2)^3)^n,$$

so h_n is a combination, with integer coefficients, of the functions

$$X_j^{(m)} := \frac{W^j}{Y^{2m}} \cdot \frac{1}{z} \quad (j \geq 0, m \geq 0)$$

with $m = 2n$ and $0 \leq j \leq 8n$. These can be integrated by a recursion on m .

Lemma 4.5. *For all integers $j \geq 0$ and $m \geq 0$, on Ω ,*

$$(4.3) \quad \frac{d}{dz} \left(\frac{W^{j+1}}{Y^{2m+1}} \right) = (j-2m) X_j^{(m)} - 8(2m+1) X_j^{(m+1)},$$

and

$$(4.4) \quad X_j^{(0)} = \frac{W^j}{z} = \sum_{k=0}^j \binom{j}{k} 2^k z^{j-2k-1}.$$

Proof. By the product rule and Lemma 3.2(b), the derivative of $W^{j+1}Y^{-2m-1}$ is $(j+1)W^jY^{-2m}/z - (2m+1)W^{j+2}Y^{-2m-2}/z$; now write $W^{j+2} = W^j(Y^2+8)$ (Lemma 3.2(a)). The second formula is the binomial theorem for $(z+2/z)^j$. (Both computations are written out in Appendix A.1.) $\square$

Read (4.3) as a formula for $X_j^{(m+1)}$:

$$(4.5) \quad X_j^{(m+1)} = \frac{1}{8(2m+1)} \left[(j-2m) X_j^{(m)} - \frac{d}{dz} \left(\frac{W^{j+1}}{Y^{2m+1}} \right) \right].$$

It expresses $X_j^{(m+1)}$ through $X_j^{(m)}$ and an exact derivative, so after m steps everything is reduced to $X_j^{(0)}$, which by (4.4) is a sum of powers of z . Each power z^{l-1} with $l \neq 0$ is the derivative of z^l/l ; the only term that is not a derivative is the one with $l = 0$, a multiple of $1/z$. Keeping track of the numbers by which one divides gives the following statement.

Proposition 4.6. *Let R be a subring of $\mathbb{C}$ containing $\frac{1}{2}$, let $m \geq 0$, and let $P \in R[W]$ be a polynomial of degree at most d . Assume that $1/l \in R$ for all integers $1 \leq l \leq d$ and $1/(2k-1) \in R$ for all integers $1 \leq k \leq m$. Then there are a number $c \in R$ and a function G on Ω such that*

$$\frac{P(W)}{Y^{2m}} \cdot \frac{1}{z} = \frac{c}{z} + G'(z) \quad \text{on } \Omega,$$

and G is a linear combination, with coefficients in R , of the functions z^l ($l \in \mathbb{Z}$, $0 < |l| \leq d$) and $W^{j+1}Y^{-(2m'+1)}$ ($0 \leq j \leq d$, $0 \leq m' < m$). Explicitly, if $P = \sum_j p_j W^j$, then

$$(4.6) \quad c = \sum_{j \text{ even}} p_j \xi_{j,m} \binom{j}{j/2} 2^{j/2}, \quad \xi_{j,m} := \prod_{i=0}^{m-1} \frac{j-2i}{8(2i+1)}.$$

Proof. By linearity it suffices to treat $P = W^j$, by induction on m . For $m = 0$ use (4.4): the exponent $l = j - 2k$ satisfies $|l| \leq j$, and the term with $l = 0$ (which occurs for even j , $k = j/2$) is $\binom{j}{j/2} 2^{j/2}/z$. The step from m to $m + 1$ is (4.5), where one divides by $8(2m + 1)$; it multiplies the constant c by $(j - 2m)/(8(2m + 1))$. $\square$

Corollary 4.7. *For every $n \geq 1$ there is a primitive pair (G, c) with $c \in \mathbb{Q}$ and $G(1+i) - G(1-i) \in i\mathbb{Q}$. Consequently the numbers u_n and v_n of (4.1) are rational, and $J_n = u_n + v_n\pi$. This proves the first assertion of Theorem 1.4(a).*

Proof. Apply Proposition 4.6 with $R = \mathbb{Q}$, $P = P_n$ and $m = 2n$; by (4.2) the result is a primitive pair. The function G is a rational combination of powers z^l and of functions $W^{j+1}Y^{-(2m'+1)}$. For these, by Lemma 3.2(d),

$$(1+i)^l - (1-i)^l = 2i \operatorname{Im}(1+i)^l \in i\mathbb{Q},$$

$$\left. \frac{W^{j+1}}{Y^{2m'+1}} \right|_{z=1+i} - \left. \frac{W^{j+1}}{Y^{2m'+1}} \right|_{z=1-i} = \frac{2^{j+1}}{(2i)^{2m'+1}} - \frac{2^{j+1}}{(-2i)^{2m'+1}} = -i(-1)^{m'} 2^{j+1-2m'} \in i\mathbb{Q}$$

(for negative l note $(1 \pm i)^{-1} = (1 \mp i)/2$). The rest is Lemma 4.3. $\square$

Example 4.8 ($n = 1$). Here $m = 2$ and $P_1(W) = (W - 3)^5(W - 2)^3 = W^8 - 21W^7 + 192W^6 - 998W^5 + 3225W^4 - 6633W^3 + 8478W^2 - 6156W + 1944$. Formula (4.6) has only three non-zero terms, because $\xi_{j,2} = j(j-2)/192$ vanishes for $j = 0, 2$:

$$c_1 = 3225 \cdot \frac{4 \cdot 2}{192} \cdot \binom{4}{2} 2^2 + 192 \cdot \frac{6 \cdot 4}{192} \cdot \binom{6}{3} 2^3 + 1 \cdot \frac{8 \cdot 6}{192} \cdot \binom{8}{4} 2^4 = 3225 + 3840 + 280 = 7345.$$

Carrying the recursion through (Appendix A.1) gives $u_1 = -\frac{23075}{2}$, so that $J_1 = -\frac{23075}{2} + \frac{7345}{2}\pi$, as stated in Example 1.6.

Remark 4.9 (a closed formula). Unwinding the recursion one finds $h_n = A_n(W)/z - E'_n(z)$ with the polynomial $A_n(W) = \sum_j p_j \xi_{j,2n} W^j$ (where $P_n = \sum_j p_j W^j$) and a function E_n that *vanishes at both end points* $1 \pm i$: each coefficient of E_n is obtained from P_n by a differential operator of order at most $2n - 1$ in W , and P_n has a zero of order $3n$ at $W = 2 = W(1 \pm i)$ (Appendix A.1). Hence

$$J_n = -i \int_{1-i}^{1+i} A_n\left(z + \frac{2}{z}\right) \frac{dz}{z} = \int_{-\pi/4}^{\pi/4} A_n(2\sqrt{2} \cos \varphi) d\varphi,$$

where the second expression is obtained by moving the path to the arc $z = \sqrt{2}e^{i\varphi}$ of the circle $|z| = \sqrt{2}$. In particular

$$c_n = \sum_{l=2n}^{4n} p_{2l} 2^{l-2n} \binom{2l}{l} \binom{l}{2n} / \binom{4n}{2n}.$$

These formulas are not needed below; we used them for an independent computation of u_n and c_n (Section 11).

5. THE DENOMINATORS OF u_n AND v_n

In this section we prove Theorem 1.4(b). The strategy is to treat one prime p at a time: for each p we find a primitive pair (G, c) in which c and $G(1+i) - G(1-i)$ have no p in the denominator, up to an explicit power of p . Since all primitive pairs give the same u_n and c_n (Lemma 4.2), this bounds the power of p in the denominators of u_n and c_n .

5.1. p -integers and partial fractions. The primitive pairs involve the irrational numbers $\sqrt{2}$ and i , so we need a notion of “no p in the denominator” for such numbers. Recall that an *algebraic integer* is a complex number that is a root of a monic polynomial with integer coefficients; the algebraic integers form a ring, and a rational number is an algebraic integer only if it is an integer [5, Chapter 2]. (Remark 5.3 shows how to do without algebraic integers.)

Definition 5.1. Let p be a prime. A complex number x is *p -integral* if Mx is an algebraic integer for some positive integer M not divisible by p . The set of p -integral numbers is denoted by R_p , and $\mathbb{Z}_{(p)} = \{a/b : a, b \in \mathbb{Z}, p \nmid b\}$.

Lemma 5.2. (a) R_p is a subring of $\mathbb{C}$. It contains $\sqrt{2}$, i , and $1/M$ for every integer M not divisible by p .

(b) $R_p \cap \mathbb{Q} = \mathbb{Z}_{(p)}$: a rational number is p -integral if and only if p does not divide its denominator.

(c) A rational number that is p -integral for every prime p is an integer.

The proof takes three lines (Appendix A.3).

Remark 5.3 (an elementary alternative). All numbers to which Definition 5.1 is applied in this paper lie in the smaller set

$$R'_p := \{a + b\sqrt{2} + ci + di\sqrt{2} : a, b, c, d \in \mathbb{Z}_{(p)}\} \subseteq R_p,$$

and a reader who prefers to avoid algebraic integers may replace R_p by R'_p throughout. The set R'_p is a ring, because $(\sqrt{2})^2 = 2$ and $i^2 = -1$; it contains $\sqrt{2}$, i , and $1/M$ for every integer M not divisible by p ; and $R'_p \cap \mathbb{Q} = \mathbb{Z}_{(p)}$, because the numbers $1, \sqrt{2}, i, i\sqrt{2}$ are linearly independent over $\mathbb{Q}$ (if $a + b\sqrt{2} + i(c + d\sqrt{2}) = 0$ with rational a, b, c, d , then $a + b\sqrt{2} = 0$ and $c + d\sqrt{2} = 0$, hence $a = b = c = d = 0$, as $\sqrt{2}$ is irrational). So Lemma 5.2 holds for R'_p with an elementary proof, and no other property of R_p is used below. (The formal proof uses R_p as in Definition 5.1.)

Remark 5.4 (background: the p -adic valuation). For a non-zero rational number x let $\text{ord}_p(x) \in \mathbb{Z}$ be the exponent of p in the factorisation of x (so $\text{ord}_3(18/5) = 2$, $\text{ord}_5(18/5) = -1$). Then $x \in \mathbb{Z}_{(p)}$ means $\text{ord}_p(x) \geq 0$, and “ $p^e x \in \mathbb{Z}_{(p)}$ ” means $\text{ord}_p(x) \geq -e$. An integer m with $1 \leq m \leq M$ satisfies $p^{\text{ord}_p(m)} \leq M$, so $\text{ord}_p(m) \leq \lfloor \log_p M \rfloor$; this is why $\text{lcm}(1, \dots, M) = \prod_p p^{\lfloor \log_p M \rfloor}$.

All functions that occur are rational functions with poles only at 0 and $\pm\sqrt{2}$. For a subring R of $\mathbb{C}$ containing $\sqrt{2}$ let $\mathcal{A}(R)$ be the set of functions on Ω of the form

$$\varphi(z) = \frac{M(z)}{z^{e_0}(z - \sqrt{2})^{e_+}(z + \sqrt{2})^{e_-}} \quad \text{with } M \in R[z] \text{ and integers } e_0, e_+, e_- \geq 0.$$

We also write e_a for the exponent that belongs to the pole $a \in \{0, \sqrt{2}, -\sqrt{2}\}$, so $e_{\sqrt{2}} = e_+$ and $e_{-\sqrt{2}} = e_-$. The set $\mathcal{A}(R)$ is closed under addition and multiplication (bring the fractions to a common denominator) and under differentiation (quotient rule), and it contains $W = (z^2 + 2)/z$ and $1/Y = z/((z - \sqrt{2})(z + \sqrt{2}))$.

Lemma 5.5 (partial fractions with coefficients in R). *Let R be a subring of $\mathbb{C}$ containing $\sqrt{2}$ and $\frac{1}{2}$. Every $\varphi \in \mathcal{A}(R)$ can be written in exactly one way as*

$$(5.1) \quad \varphi(z) = \sum_{j=0}^d t_j z^j + \sum_{a \in \{0, \sqrt{2}, -\sqrt{2}\}} \sum_{k \geq 1} \frac{q_{a,k}}{(z - a)^k} \quad (\text{finite sums}),$$

and all coefficients $t_j, q_{a,k}$ lie in R . The terms $(z - a)^{-k}$ occur only for $k \leq e_a$; in particular there are no such terms if $e_a = 0$. Moreover:

- (i) the decomposition of φ' is obtained by differentiating (5.1) term by term; in particular φ' has no terms $(z-a)^{-1}$;
- (ii) if $q_{\sqrt{2},1} = q_{-\sqrt{2},1} = 0$, then $\varphi(z) - q_{0,1}/z = G'_\varphi(z)$ on Ω , where

$$(5.2) \quad G_\varphi(z) := \sum_{j=0}^d \frac{t_j}{j+1} z^{j+1} - \sum_a \sum_{k \geq 2} \frac{q_{a,k}}{k-1} \cdot \frac{1}{(z-a)^{k-1}}.$$

The point of the hypothesis is that the distances $\sqrt{2}$ and $2\sqrt{2}$ between the poles are invertible in R ; this is true in R_p for odd p and false for $p = 2$. The proof is in Appendix A.3. The numbers by which one divides in (5.2), namely $j+1$ and $k-1$, will be called the *integration indices*.

5.2. Odd primes: the integration indices are at most $4n$.

Lemma 5.6. *Let $n \geq 1$. The decomposition (5.1) of h_n has the form*

$$(5.3) \quad h_n(z) = \sum_{j=0}^{4n-1} t_j z^j + \sum_{k=1}^{4n+1} \frac{a_k}{z^k} + \sum_{k=1}^{4n} \left(\frac{b_k}{(z-\sqrt{2})^k} + \frac{b'_k}{(z+\sqrt{2})^k} \right)$$

with coefficients t_j, a_k, b_k, b'_k in the ring $\mathbb{Z}[\sqrt{2}, \frac{1}{2}]$; in particular all coefficients are p -integral for every odd prime p . Moreover $b_1 = b'_1 = 0$ and $a_1 = c_n$. Hence (G_{h_n}, c_n) , with G_{h_n} defined by (5.2), is a primitive pair.

Proof. $h_n = N^n / (z^{4n+1}(z-\sqrt{2})^{4n}(z+\sqrt{2})^{4n})$ has a numerator with integer coefficients, so Lemma 5.5 applies with $R = \mathbb{Z}[\sqrt{2}, \frac{1}{2}]$; the polynomial part has degree $16n - (12n+1) = 4n-1$. Let (G_0, c_0) be the primitive pair of Corollary 4.7, so that $h_n = c_0/z + G'_0$. Its function G_0 is a rational combination of powers of z and of the functions $W^{j+1}Y^{-(2m'+1)}$ (Proposition 4.6), hence $G_0 \in \mathcal{A}(\mathbb{Q}(\sqrt{2}))$. By Lemma 5.5(i) the function G'_0 has no terms $(z-a)^{-1}$, so the uniqueness of the decomposition gives $b_1 = b'_1 = 0$ and $a_1 = c_0$, and $c_0 = c_n$ by definition. The last assertion is Lemma 5.5(ii). $\square$

Proposition 5.7 (odd primes). *Let p be an odd prime, $n \geq 1$, and let $e = \lfloor \log_p(4n) \rfloor$ be the largest integer with $p^e \leq 4n$. Then*

$$c_n \in \mathbb{Z}_{(p)} \quad \text{and} \quad p^e u_n \in \mathbb{Z}_{(p)}.$$

Proof. Use the primitive pair (G_{h_n}, c_n) of Lemma 5.6. First, $c_n = a_1$ is rational and p -integral, so $c_n \in \mathbb{Z}_{(p)}$ by Lemma 5.2(b). Next, in (5.2) every coefficient is a p -integral number divided by an integration index m with $1 \leq m \leq 4n$: indeed $j+1 \leq 4n$, and $k-1 \leq 4n$ for the pole 0 and $k-1 \leq 4n-1$ for the poles $\pm\sqrt{2}$. For such m the number p^e/m lies in $\mathbb{Z}_{(p)}$ (Remark 5.4). So $p^e G_{h_n}$ is a combination, with p -integral coefficients, of powers of z , $1/z$, $1/(z-\sqrt{2})$ and $1/(z+\sqrt{2})$. At $z = 1 \pm i$ these four quantities are p -integral, because

$$(5.4) \quad \frac{1}{1 \pm i} = \frac{1 \mp i}{2}, \quad \frac{1}{1 \pm i - \sqrt{2}} = -\frac{(1 \pm i + \sqrt{2})(1 \pm i)}{4}, \quad \frac{1}{1 \pm i + \sqrt{2}} = -\frac{(1 \pm i - \sqrt{2})(1 \pm i)}{4}$$

(Appendix A.3). Therefore $p^e(G_{h_n}(1+i) - G_{h_n}(1-i)) \in R_p$, and $p^e u_n = -i p^e(G_{h_n}(1+i) - G_{h_n}(1-i))$ is a rational number in R_p , hence in $\mathbb{Z}_{(p)}$. $\square$

Example 5.8 ($n = 1$). The decomposition (5.3) of $h_1 = f/z$ is

$$h_1(z) = z^3 - 21z^2 + 216z - 1460 + \frac{7345}{z} - \frac{2920}{z^2} + \frac{864}{z^3} - \frac{168}{z^4} + \frac{16}{z^5} \\ + \sum_{k=2}^4 \left(\frac{b_k}{(z - \sqrt{2})^k} + \frac{\bar{b}_k}{(z + \sqrt{2})^k} \right),$$

with $b_2 = -\frac{26633}{2} + \frac{150659}{16}\sqrt{2}$, $b_3 = \frac{47321}{4} - \frac{33461}{4}\sqrt{2}$, $b_4 = -\frac{33461}{2} + \frac{47321}{4}\sqrt{2}$, where the bar replaces $\sqrt{2}$ by $-\sqrt{2}$. We see $c_1 = 7345$ and $b_1 = 0$. The integration indices are 1, 2, 3, 4, so $\text{lcm}(1, 2, 3, 4) = 12$ clears the denominators of u_1 at the odd primes. (In fact $u_1 = -\frac{23075}{2}$ has no 3 in its denominator: D_n is a common multiple of the denominators, not always the least one.) The example also shows what goes wrong at the prime 2: the coefficient b_2 has the denominator 16.

5.3. Removable primes. By Proposition 5.7 a prime p with $\sqrt{4n} < p \leq 4n$ occurs in the denominator of u_n at most to the first power. For the removable primes of Definition 1.3 it does not occur at all.

Proposition 5.9 (removable primes). *If the prime p is removable for n , then $u_n \in \mathbb{Z}_{(p)}$.*

Proof. (Appendix A.4 comments on each step and works out an example.) A removable prime is odd, since $p^2 > 4n \geq 4$. Divide $5n$, $3n$ and $4n$ by p with remainder:

$$5n = pq_1 + \alpha_1, \quad 3n = pq_2 + \alpha_2, \quad 4n = pq_0 + \delta, \quad 0 \leq \alpha_1, \alpha_2, \delta \leq p - 1.$$

By definition, $q_0 = \lfloor 4n/p \rfloor$ is even and $\alpha_1 + \alpha_2 < \delta$. As $4n$ and pq_0 are even, δ is even, and $\delta \geq 2$.

Step 1: a p -th power splits off. By (4.2),

$$(5.5) \quad h_n = g^p \cdot \omega \quad \text{with} \quad g := \frac{(W - 3)^{q_1}(W - 2)^{q_2}}{Y^{q_0}}, \quad \omega := \frac{(W - 3)^{\alpha_1}(W - 2)^{\alpha_2}}{Y^\delta} \cdot \frac{1}{z}.$$

Step 2: the small factor ω is the derivative of a p -integral function. Apply Proposition 4.6 with $R = R_p$, $P = (W - 3)^{\alpha_1}(W - 2)^{\alpha_2}$, $d = \alpha_1 + \alpha_2$ and $m = \delta/2$. The numbers that must be invertible in R_p are 2, the integers $l \leq \alpha_1 + \alpha_2 < p$ and the odd integers $2k - 1 \leq \delta - 1 < p$: none is divisible by p . So $\omega = c_\omega/z + G'_{\text{red}}$ with $c_\omega \in R_p$ and $G_{\text{red}} \in \mathcal{A}(R_p)$. Moreover $c_\omega = 0$: in terms of z ,

$$\omega = \frac{((z - 1)(z - 2))^{\alpha_1}(z^2 - 2z + 2)^{\alpha_2} z^{\delta - \alpha_1 - \alpha_2 - 1}}{(z^2 - 2)^\delta}$$

has no power of z in its denominator, because $\delta - \alpha_1 - \alpha_2 - 1 \geq 0$. So it belongs to $\mathcal{A}(R_p)$ with $e_0 = 0$, and its decomposition (5.1) has no term $1/z$ (Lemma 5.5); and G'_{red} has no term $1/z$ either (Lemma 5.5(i)). Hence $c_\omega/z = \omega - G'_{\text{red}}$ has no term $1/z$, which by the uniqueness of the decomposition means $c_\omega = 0$. Thus $\omega = G'_{\text{red}}$. This is where the inequality $\alpha_1 + \alpha_2 < \delta$ is used; the parity of q_0 was used to make δ even, so that Proposition 4.6 applies.

Step 3: integration by parts. Put $H := g^p G_{\text{red}}$ and $\eta := g^{p-1} g' G_{\text{red}}$. Both belong to $\mathcal{A}(R_p)$, and by the product rule $H' = p g^{p-1} g' G_{\text{red}} + g^p G'_{\text{red}}$, that is

$$(5.6) \quad h_n = H' - p \eta.$$

Step 4: compare coefficients. Let $t_j^H, q_{a,k}^H$ and $t_j^\eta, q_{a,k}^\eta$ be the coefficients of H and η in (5.1); they are p -integral by Lemma 5.5. By (5.6), Lemma 5.5(i) and the uniqueness of the decomposition, the coefficients of h_n in (5.3) are

$$t_j = (j + 1) t_{j+1}^H - p t_j^\eta, \quad q_{a,k} = -(k - 1) q_{a,k-1}^H - p q_{a,k}^\eta \quad (k \geq 2).$$

Dividing by the integration indices, the coefficients of the primitive G_{h_n} in (5.2) are

$$(5.7) \quad \frac{t_j}{j+1} = t_{j+1}^H - \frac{p}{j+1} t_j^\eta, \quad -\frac{q_{a,k}}{k-1} = q_{a,k-1}^H + \frac{p}{k-1} q_{a,k}^\eta.$$

The indices $j+1$ and $k-1$ are at most $4n < p^2$, so they are divisible by p at most once and $p/(j+1)$, $p/(k-1)$ lie in $\mathbb{Z}_{(p)}$. Hence *every* coefficient of G_{h_n} is p -integral, and as in the proof of Proposition 5.7 we conclude $u_n = -i(G_{h_n}(1+i) - G_{h_n}(1-i)) \in R_p \cap \mathbb{Q} = \mathbb{Z}_{(p)}$. $\square$

In words: by (5.6) the function h_n is, modulo p , the derivative of the p -integral function H , and the derivative of t^m is mt^{m-1} ; so a coefficient of h_n whose integration index is divisible by p is itself divisible by p , and the division by the index costs nothing.

Example 5.10 ($n = 2$, $p = 3$). Here $4n = 8 < 9$, $\lfloor 8/3 \rfloor = 2$ is even, and $(10 \bmod 3) + (6 \bmod 3) = 1 + 0 < 2 = 8 \bmod 3$: the prime 3 is removable, $\Phi_2 = 3$. We have $10 = 3 \cdot 3 + 1$, $6 = 3 \cdot 2 + 0$, $8 = 3 \cdot 2 + 2$, so

$$g = \frac{(W-3)^3(W-2)^2}{Y^2}, \quad \omega = \frac{W-3}{Y^2} \cdot \frac{1}{z} = \frac{(z-1)(z-2)}{(z^2-2)^2} = G'_{\text{red}},$$

$$G_{\text{red}} = \frac{3W-8}{8Y} = \frac{3z^2-8z+6}{8(z^2-2)}.$$

The only denominator in G_{red} is 8. In the decomposition (5.3) of h_2 the integration indices divisible by 3 are 3 and 6, and the corresponding coefficients are indeed multiples of 3: for instance $t_2 = -998316 = 3 \cdot (-332772)$, $t_5 = 873 = 3 \cdot 291$, $a_4 = -7986528 = 3 \cdot (-2662176)$, $a_7 = 55872 = 3 \cdot 18624$. Accordingly $u_2 = -\frac{4338189943}{14}$ has no 3 in its denominator, although $\text{lcm}(1, \dots, 8) = 840$ does contain the factor 3; and $D_2 = 840/3 = 280$. The full computation is in Appendix A.4.

Remark 5.11. The case $k = 1$ of the comparison in Step 4 reads $c_n = a_1 = -p q_{0,1}^\eta$. So every removable prime divides the integer c_n ; for example $3 \mid c_2 = 197269821$ and $7 \mid c_5$. This is not used in the sequel.

5.4. The prime 2.

Proposition 5.12 (the prime 2). *Let $n \geq 1$ and $L = \lfloor \log_2(4n) \rfloor$. Then*

$$c_n \in \mathbb{Z}_{(2)} \quad \text{and} \quad 2^L u_n \in \mathbb{Z}_{(2)}.$$

So the prime 2 costs no more than what $\text{lcm}(1, \dots, 4n)$ already contains. This cannot be proved with the decomposition (5.3): the distances $\sqrt{2}$ and $2\sqrt{2}$ between the poles are not invertible in R_2 , and the individual coefficients have large powers of 2 in their denominators (for $n = 2$ the denominator of b_2 is 1024, whereas that of u_2 is 14). The powers of 2 cancel in the sum. To make the cancellation visible we change the coordinate.

The coordinate w . Let

$$\varepsilon := \sqrt{2} - 1, \quad \varkappa := 1 + \varepsilon^2 = 2\sqrt{2}\varepsilon.$$

Note that $\varepsilon(\sqrt{2}+1) = 1$, so ε and $1/\varepsilon = \sqrt{2}+1$ both lie in $\mathbb{Z}[\sqrt{2}]$ (in particular they are algebraic integers), and that $\varepsilon^2 + 2\varepsilon = 1$. Put

$$(5.8) \quad w = w(z) := i\varepsilon \frac{\sqrt{2} + z}{\sqrt{2} - z}, \quad \text{equivalently} \quad z = \sqrt{2} \frac{w - i\varepsilon}{w + i\varepsilon}.$$

This Möbius map sends $z = -\sqrt{2}$, $\sqrt{2}$, 0 , ∞ to $w = 0$, ∞ , $i\varepsilon$, $-i\varepsilon$, the circle $|z| = \sqrt{2}$ to the real axis, and the end points $1-i$, $1+i$ to $w = 1$, -1 .

Lemma 5.13. For $z \in \Omega$ and $w = w(z)$ one has $w \neq 0$, $w^2 + \varepsilon^2 \neq 0$, and

- (a) $f(z) = \frac{1}{128\varepsilon} \cdot \frac{(1 + \varepsilon^2 w^2)^5 (1 - w^2)^3}{w^4 (w^2 + \varepsilon^2)^4}$;
- (b) $\frac{1}{z} = \frac{2i\varepsilon}{w^2 + \varepsilon^2} \cdot \frac{dw}{dz}$;
- (c) $w(1+i) = -1$ and $w(1-i) = 1$.

Consequently

$$(5.9) \quad h_n(z) = \frac{2i\varepsilon}{(128\varepsilon)^n} k_n(w(z)) \frac{dw}{dz}, \quad k_n(w) := \frac{(1 + \varepsilon^2 w^2)^{5n} (1 - w^2)^{3n}}{w^{4n} (w^2 + \varepsilon^2)^{4n+1}}.$$

The proof is a computation with the four factors of f (Appendix A.5). The gain is that k_n is an *even* function of w with poles only at 0 and $\pm i\varepsilon$, to be “integrated from -1 to 1 ”.

Data of a function. We do the bookkeeping of primitives directly in the variable w .

Definition 5.14. Let φ be a function on $\mathbb{C} \setminus \{0, i\varepsilon, -i\varepsilon\}$. We say that φ has data $(U, V) \in \mathbb{C}^2$ if there is a holomorphic function F on $\mathbb{C} \setminus \{0, i\varepsilon, -i\varepsilon\}$ such that

$$F'(w) = \varphi(w) - \frac{V}{w^2 + \varepsilon^2} \quad \text{and} \quad F(1) - F(-1) = U.$$

Informally: “ $\int_{-1}^1 \varphi(w) dw = U + V\vartheta$, where $\vartheta = \int_{-1}^1 dw/(w^2 + \varepsilon^2)$ ” (the first integral along a path from -1 to 1 that avoids the point 0, where φ may have a pole), but the logarithmic integral ϑ is never evaluated. Data are additive: if φ_1, φ_2 have data $(U_1, V_1), (U_2, V_2)$, then $\varphi_1 + \varphi_2$ has data $(U_1 + U_2, V_1 + V_2)$, and $b\varphi_1$ has data (bU_1, bV_1) for every complex number b .

Lemma 5.15 (transfer to z). If k_n has data (U, V) , then

$$(5.10) \quad u_n = -\frac{2\varepsilon}{(128\varepsilon)^n} U, \quad c_n = \frac{1}{(128\varepsilon)^n} V.$$

Proof. With F as in Definition 5.14 put $G(z) := 2i\varepsilon(128\varepsilon)^{-n} F(w(z))$ and $c := (128\varepsilon)^{-n} V$. By the chain rule, (5.9) and Lemma 5.13(b),

$$G'(z) = \frac{2i\varepsilon}{(128\varepsilon)^n} \left(k_n(w) - \frac{V}{w^2 + \varepsilon^2} \right) \frac{dw}{dz} = h_n(z) - \frac{c}{z},$$

so (G, c) is a primitive pair. By Lemma 5.13(c), $G(1+i) - G(1-i) = 2i\varepsilon(128\varepsilon)^{-n} (F(-1) - F(1))$; multiply by $-i$ and use Lemma 4.2. $\square$

Measuring powers of 2. The data will be numbers of the field $\mathbb{Q}(\sqrt{2})$, in which $2 = (\sqrt{2})^2$. We measure divisibility by powers of $\sqrt{2}$.

Definition 5.16. For $k \in \mathbb{Z}$ let $\mathfrak{L}_k := \{(\sqrt{2})^k y : y \in R_2\}$. If $x \in \mathfrak{L}_k$ we say that x has level at least k .

So $\mathfrak{L}_0 = R_2$, and level $2k$ for a rational number means “divisible by 2^k ”. Levels add under multiplication, and a sum of numbers of level $\geq k$ has level $\geq k$. The facts we need are collected in the following lemma (proofs in Appendix A.5); $s_2(a)$ denotes the number of digits 1 in the binary expansion of a .

Lemma 5.17. (a) $\mathfrak{L}_j \cdot \mathfrak{L}_k \subseteq \mathfrak{L}_{j+k}$, $\mathfrak{L}_k + \mathfrak{L}_k \subseteq \mathfrak{L}_k$, and $\mathfrak{L}_k \subseteq \mathfrak{L}_j$ for $j \leq k$.

(b) $2^k \in \mathfrak{L}_{2k}$ for $k \in \mathbb{Z}$. If m is an odd integer, m and $1/m$ lie in $\mathfrak{L}_0$. For $k \in \mathbb{Z}$: $\varepsilon^k \in \mathfrak{L}_0$ and $\varepsilon^k \in \mathfrak{L}_{3k}$.

(c) (Legendre) $a!$ equals $2^{a-s_2(a)}$ times an odd integer. (Kummer) $\binom{2j}{j}$ equals $2^{s_2(j)}$ times an odd integer. Moreover $2s_2(a) \leq a + 1$ for $a \geq 0$, and $s_2(a) \leq \lfloor \log_2 a \rfloor + 1$ for $a \geq 1$.

(d) If a rational number x lies in $\mathfrak{L}_{-2L}$, then $2^L x \in \mathbb{Z}_{(2)}$.

The building blocks. Introduce

$$x := \frac{1 - w^2}{\varkappa}, \quad y := 1 - x = \frac{w^2 + \varepsilon^2}{\varkappa}.$$

The third factor of k_n is also a combination of x and y : since $\varepsilon^2 + 2\varepsilon = 1$,

$$(5.11) \quad 1 + \varepsilon^2 w^2 = \varkappa(y + 2\varepsilon x).$$

Inserting this in (5.9) and expanding $(y + 2\varepsilon x)^{5n}$ by the binomial theorem,

$$(5.12) \quad k_n(w) = \varkappa^{4n-1} \sum_{i=0}^{5n} \binom{5n}{i} (2\varepsilon)^i \frac{x^{3n+i} y^{n-1-i}}{w^{4n}}.$$

(From here to the end of the proof of Proposition 5.12 the letter i denotes a summation index; the imaginary unit does not occur.) For $i \leq n-1$ the exponent of y is non-negative and $y^{n-1-i} = (1-x)^{n-1-i}$ is a polynomial in x . For $i \geq n$ it is negative; then we write $x^{3n+i} = (1-y)^{3n+i}$ and expand in powers of y . In both cases we arrive at two kinds of “atoms”.

Lemma 5.18 (the atoms). *Let $m \geq 0$ be an integer.*

(a) *For every integer $a \geq 0$ the function $x^a w^{-2m}$ has data $(B_{m,a}, 0)$ with*

$$B_{m,a} = \frac{2^{a+1} a!}{\varkappa^a \prod_{j=0}^a (2j+1-2m)} \in \mathfrak{L}_{a+2-2s_2(a)}.$$

(b) *For every integer $k \geq 1$ the function $y^{-k} w^{-2m}$ has data (U, V) with $U, V \in \mathfrak{L}_{4-k}$.*

For (a) one expands $(1-w^2)^a$ and integrates the even powers of w ; the alternating sum that arises has a closed form, in which the factorial $a!$ supplies many factors 2. For (b) one reduces to $m=0$ by partial fractions in w^2 and uses the classical recursion for $\int (w^2 + \varepsilon^2)^{-j} dw$, whose coefficients are central binomial coefficients. All details are in Appendix A.5.

Proof of Proposition 5.12. We show that k_n has data (U, V) with

$$(5.13) \quad U \in \mathfrak{L}_{14n-2-2L}, \quad V \in \mathfrak{L}_{14n}.$$

Then (5.10) gives the claim: $(128\varepsilon)^{-n} = 2^{-7n} \varepsilon^{-n} \in \mathfrak{L}_{-14n}$ and $2\varepsilon \in \mathfrak{L}_2$, so $c_n \in \mathfrak{L}_0 = R_2$ and $u_n \in \mathfrak{L}_{-2L}$; as both are rational, $c_n \in \mathbb{Z}_{(2)}$ and $2^L u_n \in \mathbb{Z}_{(2)}$ by Lemmas 5.2(b) and 5.17(d).

Consider the i -th term $T_i := x^{3n+i} y^{n-1-i} w^{-4n}$ of (5.12); we use Lemma 5.18 with $m=2n$.

Case $i \leq n-1$. Then $T_i = \sum_b \binom{n-1-i}{b} (-1)^b x^{3n+i+b} w^{-4n}$ is a combination with integer coefficients of atoms $x^a w^{-4n}$ with $3n \leq a \leq 4n-1$. For such a we have $s_2(a) \leq \lfloor \log_2 a \rfloor + 1 \leq L+1$, so $B_{2n,a}$ has level at least $a+2-2s_2(a) \geq 3n-2L$. Hence T_i has data $(U_i, 0)$ with $U_i \in \mathfrak{L}_{3n-2L}$.

Case $i \geq n$. Put $j := i-n+1 \geq 1$, so that $T_i = x^{3n+i} y^{-j} w^{-4n} = \sum_l \binom{3n+i}{l} (-1)^l y^{l-j} w^{-4n}$. If $l \geq j$, then $y^{l-j} = (1-x)^{l-j}$ is a polynomial in x and we get atoms $x^b w^{-4n}$ with $b \geq 0$, of level at least $b+2-2s_2(b) \geq 1$. If $l < j$ we get atoms $y^{-k} w^{-4n}$ with $1 \leq k = j-l \leq j$, whose data have level at least $4-k \geq 4-j$. Hence T_i has data (U_i, V_i) with $U_i \in \mathfrak{L}_{\min(1, 4-j)}$ and $V_i \in \mathfrak{L}_{4-j}$.

Now multiply by $\binom{5n}{i} (2\varepsilon)^i \in \mathfrak{L}_{2i}$ and by $\varkappa^{4n-1} \in \mathfrak{L}_{12n-3}$ and add up. Table 5 shows the resulting levels: the contributions to U have level at least $14n-2-2L$ and those to V have level at least $14n$. This proves (5.13). $\square$

Example 5.19 ($n=1$). Here $L=2$, $k_1 = \varkappa^3 \sum_{i=0}^5 \binom{5}{i} (2\varepsilon)^i x^{3+i} y^{-i} w^{-4}$, and the six terms contribute to the data (U, V) of k_1 as follows:

terms of (5.12)	contribution to U : level	contribution to V : level
$0 \leq i \leq n-1$	$\geq (12n-3) + 2i + (3n-2L)$	none
$n \leq i \leq 5n$, atoms $x^b w^{-4n}$	$\geq (12n-3) + 2i + 1 \geq 14n-2$	none
$n \leq i \leq 5n$, atoms $y^{-k} w^{-4n}$	$\geq (12n-3) + 2i + (4-j) = 13n+i \geq 14n$	$\geq 13n+i \geq 14n$

TABLE 5. Levels (powers of $\sqrt{2}$) of the three groups of terms in the proof of Proposition 5.12, after multiplication by $\varkappa^{4n-1} \binom{5n}{i} (2\varepsilon)^i$; here $j = i - n + 1$. All entries of the middle column are $\geq 14n - 2 - 2L$.

i	0	1	2	3	4	5	sum
contribution to U	$\frac{32}{3}$	$\frac{1280}{3}$	10240	$\frac{240640}{3}$	281600	$\frac{1097728}{3}$	738400
contribution to V/ε	0	640	12800	102400	358400	465920	940160

All contributions to U are divisible by 2^5 , so they have level at least 10, and all contributions to V are ε times an integer divisible by 2^7 , so they have level at least 14. This is in accordance with (5.13), which requires the levels $14n - 2 - 2L = 8$ and $14n = 14$. The columns $i = 0$ and $i = 1$ are computed by hand at the end of Appendix A.5, where it is also explained why every entry is a rational number. By (5.10), $u_1 = -2\varepsilon \cdot 738400/(128\varepsilon) = -\frac{23075}{2}$ and $c_1 = 940160\varepsilon/(128\varepsilon) = 7345$, in agreement with Example 4.8. Compare with Example 5.8, where the coefficient b_2 had the denominator 16.

Remark 5.20 (the classical formulation). In the original version of the argument one evaluates $\vartheta = \int_{-1}^1 dw/(w^2 + \varepsilon^2) = \frac{2}{\varepsilon} \arctan \frac{1}{\varepsilon} = \frac{3\pi}{4\varepsilon}$, integrates k_n from -1 to 1 along the real w -axis with a small detour around the pole $w = 0$ (in the z -plane: the long arc of the circle $|z| = \sqrt{2}$ from $1+i$ to $1-i$ with a detour around $-\sqrt{2}$; the side does not matter, because k_n is even and so has no residue at 0), and obtains $-2\varepsilon(128\varepsilon)^{-n}(U + V\vartheta) = u_n - 3v_n\pi$, since $\log z$ increases by $\frac{3}{2}\pi i$ along this path. The comparison of the coefficients of π requires $\pi \notin \mathbb{Q}(\sqrt{2})$. With primitive pairs the comparison is replaced by Lemma 4.2 and the number ϑ never appears.

5.5. Integrality.

Theorem 5.21. *For every $n \geq 1$ the number c_n is an integer, and $D_n u_n$ and $D_n v_n$ are integers. This proves Theorem 1.4(b) and the integrality of c_n in Theorem 1.4(a).*

Proof. By Lemma 5.2(c) it suffices to show that the three rational numbers are p -integral for every prime p .

The number c_n lies in $\mathbb{Z}_{(p)}$ for odd p (Proposition 5.7) and in $\mathbb{Z}_{(2)}$ (Proposition 5.12).

The number $D_n u_n$. Let $L_p = \lfloor \log_p(4n) \rfloor$, so that p^{L_p} is the exact power of p dividing $\text{lcm}(1, \dots, 4n)$. If $p = 2$: the prime 2 is not removable (it violates $p^2 > 4n$), so 2^{L_2} divides D_n , and $2^{L_2} u_n \in \mathbb{Z}_{(2)}$ by Proposition 5.12. If p is odd and not removable, then again p^{L_p} divides D_n and $p^{L_p} u_n \in \mathbb{Z}_{(p)}$ by Proposition 5.7 (for $p > 4n$ this reads $u_n \in \mathbb{Z}_{(p)}$). If p is removable, $u_n \in \mathbb{Z}_{(p)}$ by Proposition 5.9. In all cases $D_n u_n \in \mathbb{Z}_{(p)}$.

The number $D_n v_n = \frac{1}{2} D_n c_n$. Since $4 \mid \text{lcm}(1, \dots, 4n)$ and Φ_n is odd, D_n is even, and c_n is an integer. $\square$

6. THE GROWTH OF v_n

We now turn to the sizes. Recall $v_n = \frac{1}{2} c_n$, where c_n is the constant of any primitive pair.

Lemma 6.1. *c_n is the coefficient of z^{4n} in the power series expansion of $N(z)^n (z^2 - 2)^{-4n}$ at $z = 0$.*

Proof. Let $T(z)$ be the Taylor polynomial of degree $4n$ of the power series $S(z) = N(z)^n(z^2 - 2)^{-4n}$. Then $N^n - (z^2 - 2)^{4n}T = (z^2 - 2)^{4n}(S - T)$ is a polynomial, and it is divisible by z^{4n+1} because the power series $S - T$ is; say it equals $z^{4n+1}Q(z)$. Hence

$$h_n(z) = \frac{N(z)^n}{z^{4n+1}(z^2 - 2)^{4n}} = \frac{T(z)}{z^{4n+1}} + \frac{Q(z)}{(z^2 - 2)^{4n}}.$$

The first summand is a sum of powers z^{-k} , $1 \leq k \leq 4n + 1$, and its coefficient of z^{-1} is the top coefficient of T ; the second summand has no power of z in its denominator ($e_0 = 0$), so its decomposition (5.1) contains no powers z^{-k} (Lemma 5.5). By the uniqueness of the decomposition (5.3), a_1 is the coefficient of z^{4n} in S ; and $a_1 = c_n$ by Lemma 5.6. $\square$

Lemma 6.1 is the second assertion of Theorem 1.4(a). The power series $S(z)$ has coefficients of both signs, but after the substitution $z \mapsto -z$ all signs become positive. Put

$$\tilde{N}(z) := N(-z) = ((z+1)(z+2))^5(z^2+2z+2)^3, \quad \tilde{P}(z) := \frac{\tilde{N}(z)}{(2-z^2)^4} = \sum_{k \geq 0} \chi_k z^k.$$

Since $1/(2-z^2) = \sum_{k \geq 0} 2^{-k-1}z^{2k}$, all coefficients χ_k are non-negative, and the series converges for $|z| < \sqrt{2}$. Note that $\tilde{P}(x)/x^4 = f(-x)$.

Lemma 6.2. (a) c_n is the coefficient of z^{4n} in $\tilde{P}(z)^n$.

(b) $c_{m+n} \geq c_m c_n$ for all $m, n \geq 1$, and $c_n \geq c_1^n = 7345^n > 0$.

(c) $c_n \leq f(-x)^n$ for every real x with $0 < x < \sqrt{2}$.

Proof. (a) $S(-z) = \tilde{N}(z)^n(z^2 - 2)^{-4n} = \tilde{P}(z)^n$, and the coefficient of the even power z^{4n} is the same in $S(z)$ and $S(-z)$. (b) The coefficient of $z^{4(m+n)}$ in $\tilde{P}^m \cdot \tilde{P}^n$ is a sum of non-negative products, one of which is $c_m c_n$. (c) All terms of the convergent series $\tilde{P}(x)^n = \sum_k [\text{coefficient of } z^k \text{ in } \tilde{P}^n] x^k$ are non-negative, so the single term $c_n x^{4n}$ is at most the sum: $c_n \leq (\tilde{P}(x)/x^4)^n$. $\square$

Proposition 6.3 (growth). *The limit $r = \lim_{n \rightarrow \infty} \frac{1}{n} \log c_n = \lim_{n \rightarrow \infty} \frac{1}{n} \log v_n$ exists, and*

$$8.9 < \log 7345 \leq r \leq \log f(-x) \quad \text{for every } 0 < x < \sqrt{2}.$$

In particular $r \leq \log f(-0.3913) < 10.540357$. This proves Theorem 1.4(c) and the positivity of c_n .

Proof. By Lemma 6.2(b),(c) the sequence $\log c_n$ is superadditive and $\frac{1}{n} \log c_n$ is bounded above. By Fekete's lemma (Appendix A.6) $\frac{1}{n} \log c_n$ converges to its supremum r , which is at least $\log c_1$ and at most $\log f(-x)$ for every x . For the rational point $x = \frac{3913}{10000}$ one computes $f(-x) = 37810.9916 \dots < 37811 < e^{10.540357}$ (Appendix A.6). $\square$

Remark 6.4 (saddle-point heuristics; the exact value of r). Why $x = 0.3913$? On the interval $(0, \sqrt{2})$ the function $x \mapsto f(-x)$ tends to $+\infty$ at both ends and has exactly one critical point, $x_0 = 0.3912692430 \dots$, where it takes its minimal value

$$\lambda_3 = f(-x_0) = 37810.99119500591 \dots, \quad \log \lambda_3 = 10.5403551116 \dots$$

So the best bound obtainable from Lemma 6.2(c) is $r \leq \log \lambda_3$, and 0.3913 is a rational point next to x_0 . In fact $r = \log \lambda_3$; this is not proved in the present paper and it is not needed, but the reason is easy to explain. It is the general principle behind the *saddle-point method*: by Cauchy's formula $c_n = \frac{1}{2\pi i} \oint f(z)^n dz/z$ over any circle $|z| = x < \sqrt{2}$, and on this circle $|f|$ is largest at the point $z = -x$. Moving the circle to the radius x_0 , where $f(-x)$ is smallest, makes the integrand as small as possible; at $z = -x_0$ the function $|f|$ has a *saddle point* (a

minimum along the real axis and a maximum along the circle), and the integral is dominated by a neighbourhood of this point, which gives $c_n = \lambda_3^n n^{-1/2}(\text{const} + o(1))$. The original draft derives this refinement from a local limit theorem of probability theory; it agrees with the exact data: $\log v_n - n \log \lambda_3 + \frac{1}{2} \log n = -2.3230$ and -2.3229 for $n = 200$ and 400 . We do not need it: Hata's lemma requires the existence of the limit r , which Fekete's lemma provides, and the bound $1 + \sigma/\tau$ is increasing in r , so an upper bound for r suffices. The number λ_3 is the real root of $2^{20}\lambda^3 - 39647694848\lambda^2 - 115523712\lambda - 3^3 \cdot 5^5 = 0$ (Appendix B).

7. THE DECAY OF J_n

On the path of integration $|f|$ is small, and $|J_n| \leq 2 \max |f|^n$ holds trivially (the segment has length 2 and $|z| \geq 1$ on it). On the segment the maximum of $|f|$ is $e^{-6.0709\dots}$, attained at $z = 1 \pm 0.5428\dots i$. The integral is much smaller than this suggests, and to see it one moves the path (Figure 2 compares the size of f on the two paths).

Let

$$z_s := \frac{19787}{16384} + \frac{32371}{65536}i = 1.20770\dots + 0.49394\dots i, \quad \rho := \frac{1458809}{10^9} = 0.001458809,$$

and let Γ be the polygon with the vertices $1 - i, \bar{z}_s, 1, z_s, 1 + i$ (Figure 1).

Proposition 7.1. $|f(z)| \leq \rho$ for every point z of the polygon Γ .

Proof (computer-assisted). Since $f(\bar{z}) = \overline{f(z)}$, it suffices to consider the two segments from 1 to z_s and from z_s to $1 + i$. For $z = x + iy$ with real x, y ,

$$(7.1) \quad |f(z)|^2 = \frac{\text{Num}(x, y)}{\text{Den}(x, y)}$$

with two explicit polynomials with integer coefficients, $\text{Den} > 0$ on Ω (Appendix A.7). On a segment, x and y are polynomials of degree 1 in the parameter $t \in [0, 1]$, so the claim $|f| \leq \rho$ becomes

$$Q(t) := \rho^2 \text{Den}(x(t), y(t)) - \text{Num}(x(t), y(t)) \geq 0 \quad (0 \leq t \leq 1),$$

where Q is a polynomial of degree 32 with rational coefficients. A polynomial inequality of this kind can be *certified*: write $Q(t_0 + \ell s)$ in the form $\sum_{j=0}^{32} \beta_j s^j (1-s)^{32-j}$; if all β_j are ≥ 0 , then $Q \geq 0$ on $[t_0, t_0 + \ell]$, because every term is ≥ 0 for $0 \leq s \leq 1$. For the segment from 1 to z_s all 33 coefficients β_j for the whole interval $[0, 1]$ are positive; for the segment from z_s to $1 + i$ this is the case on each of the halves $[0, \frac{1}{2}]$ and $[\frac{1}{2}, 1]$. The coefficients are computed in exact rational arithmetic. They are listed in the file `ContourCert.lean` of the accompanying material (Section 1.7), where Lean verifies the three polynomial identities and the signs, and they are recomputed independently by the program of Section 11; see Appendix A.7 for more details. $\square$

Remark 7.2 (background: Bernstein form). The polynomials $\binom{d}{j} s^j (1-s)^{d-j}$, $0 \leq j \leq d$, are the *Bernstein polynomials* of degree d ; they are non-negative on $[0, 1]$ and form a basis of the polynomials of degree at most d . A polynomial whose coefficients in this basis are non-negative is therefore visibly non-negative on $[0, 1]$. For a polynomial that is strictly positive on an interval this always succeeds after the interval has been cut into sufficiently short pieces; here one cut suffices. The certificate consists of 3×33 rational numbers; checking it requires only the expansion of a polynomial identity and the inspection of signs (Appendix A.7).

Proposition 7.3 (decay). *For every $n \geq 1$,*

$$|J_n| \leq 4\rho^n < 4e^{-6.530134n}.$$

This proves Theorem 1.4(d). (For $n = 1$: $|J_1| = 0.00098$ and $4\rho = 0.0058$.)

Proof. Let (G, c) be a primitive pair. As in the proof of Lemma 4.3, $F := G + c \operatorname{Log}$ is a primitive of h_n on $\{\operatorname{Re} z > 0\} \setminus \{\sqrt{2}\}$. Every point of Γ has real part between 1 and $\frac{19787}{16384} < \frac{5}{4} < \sqrt{2}$, so Γ lies in this set, and for each of its four segments $[a, b]$ the fundamental theorem of calculus gives $\int_a^b h_n dz = F(b) - F(a)$. The sum telescopes:

$$\int_{\Gamma} h_n(z) dz = F(1+i) - F(1-i) = \int_{1-i}^{1+i} h_n(z) dz, \quad \text{so} \quad J_n = -i \int_{\Gamma} f(z)^n \frac{dz}{z}.$$

(No theorem about deformation of contours is needed: the same primitive serves both paths.) On Γ we have $|f| \leq \rho$ and $|z| \geq \operatorname{Re} z \geq 1$, and each of the four segments has length less than 1. Hence $|J_n| \leq 4\rho^n$. Finally $\rho < e^{-6.530134}$ (for the lengths and for this inequality see Appendix A.7). $\square$

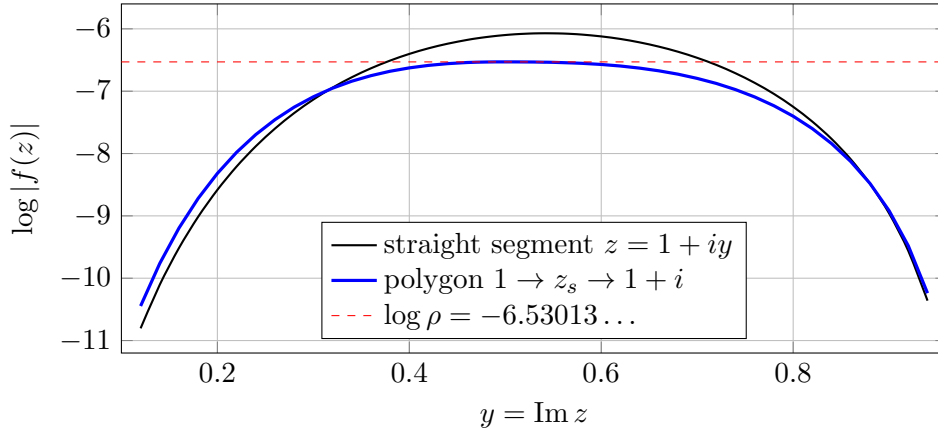


FIGURE 2. The size of f along the upper half of the two paths, as a function of $y = \operatorname{Im} z$ (the lower half is the mirror image; at the zeros $z = 1$ and $z = 1 + i$ of f the curves go to $-\infty$). On the straight segment the maximum of $\log |f|$ is -6.0709 ; on the polygon $\log |f|$ stays below $\log \rho$ (Proposition 7.1), and it comes close to this value only near the vertex z_s , at $y = 0.4939$.

Remark 7.4 (why this polygon). Think of $|f|$ as the height of a landscape over the z -plane. It has no local maxima away from the poles, so the lowest path from $1 - i$ to $1 + i$ crosses the “mountain range” at a pass, that is at a saddle point, where $f' = 0$. In the coordinate W the critical points of f other than its zeros are the roots of $4W^3 + W^2 - 88W + 152 = 0$: the real root $W = -5.5028\dots$ corresponds to $z = -x_0$ (and to $-2/x_0$), which governs the growth of v_n (Remark 6.4), and the complex roots $W = 2.62641\dots \pm 0.08629\dots i$ correspond to four saddle points $z_1, \bar{z}_1, 2/z_1, 2/\bar{z}_1$ with

$$z_1 = 1.20770992\dots + 0.49394723\dots i, \quad |f(z_1)| = e^{-6.53013614\dots} = 0.00145880722\dots$$

The vertex z_s of our polygon is z_1 rounded to dyadic rationals, $|z_s - z_1| < 10^{-5}$, and ρ exceeds $|f(z_1)|$ by a factor $1 + 1.2 \cdot 10^{-6}$ only. The exact data suggest that this is optimal: $-\frac{1}{n} \log |J_n|$ approaches 6.53 from above (Table 7).

8. THE MULTIPLIER D_n

It remains to determine the size of $D_n = \operatorname{lcm}(1, \dots, 4n)/\Phi_n$. First we describe the removable primes more conveniently.

Lemma 8.1 (windows). *Let $n \geq 1$ and let p be a prime with $p^2 > 4n$. Then p is removable for n if and only if*

$$\left\{\frac{n}{p}\right\} \in \left[\frac{1}{5}, \frac{1}{4}\right) \cup \left[\frac{2}{3}, \frac{3}{4}\right).$$

The proof is a case distinction over ten subintervals of $[0, 1)$ (Appendix A.8). A prime $p \leq 4n$ has $n/p \geq \frac{1}{4}$, so the lemma says that the primes dividing Φ_n are the primes $p > \sqrt{4n}$ in one of the windows

$$(8.1) \quad \left(\frac{4n}{4k+3}, \frac{3n}{3k+2}\right] \quad (k = 0, 1, 2, \dots) \quad \text{or} \quad \left(\frac{4n}{4k+1}, \frac{5n}{5k+1}\right] \quad (k = 1, 2, 3, \dots),$$

which correspond to $k + \frac{2}{3} \leq \frac{n}{p} < k + \frac{3}{4}$ and to $k + \frac{1}{5} \leq \frac{n}{p} < k + \frac{1}{4}$. For example, for $n = 21$ the windows are $(28, 31.5]$, $(16.8, 17.5]$, $(12, 12.6]$, $\dots$, and the primes in them are 29, 31 and 17: $\Phi_{21} = 17 \cdot 29 \cdot 31$.

Remark 8.2 (background: Chebyshev's functions and the prime number theorem). For real $x \geq 1$ let

$$\theta(x) = \sum_{p \leq x} \log p, \quad \psi(x) = \sum_{p^k \leq x} \log p = \log \text{lcm}(1, 2, \dots, \lfloor x \rfloor),$$

where p runs through the primes and $k \geq 1$. The *prime number theorem* states that $\theta(x)/x \rightarrow 1$ as $x \rightarrow \infty$. Since $0 \leq \psi(x) - \theta(x) \leq \sqrt{x} \log x$ (Appendix A.8), also $\psi(x)/x \rightarrow 1$; in particular $\text{lcm}(1, \dots, N) = e^{N+o(N)}$. The sum of $\log p$ over the primes in an interval $(\alpha n, \beta n]$ is $\theta(\beta n) - \theta(\alpha n) = (\beta - \alpha)n + o(n)$. For proofs of the prime number theorem and of the equivalence of its various forms see [1, Chapters 4 and 13]. (Chebyshev's function is always written with its argument, $\theta(x)$; it should not be confused with the irrational number θ of Definition 1.1 and Lemma 1.2.)

For an integer $K \geq 1$ let $\Phi_n^{(K)}$ be the product of the primes in the windows (8.1) with $k < K$, and

$$(8.2) \quad D_n^{(K)} := \frac{\text{lcm}(1, \dots, 4n)}{\Phi_n^{(K)}}, \quad \varpi_K := \sum_{k=0}^{K-1} \frac{1}{(3k+2)(4k+3)} + \sum_{k=1}^{K-1} \frac{1}{(4k+1)(5k+1)}.$$

The terms of ϖ_K are the lengths of the windows divided by n : $\frac{3}{3k+2} - \frac{4}{4k+3} = \frac{1}{(3k+2)(4k+3)}$ and $\frac{5}{5k+1} - \frac{4}{4k+1} = \frac{1}{(4k+1)(5k+1)}$.

Proposition 8.3 (the multiplier). *Assume the prime number theorem.*

- (a) *For every fixed $K \geq 1$: $\lim_{n \rightarrow \infty} \frac{1}{n} \log D_n^{(K)} = 4 - \varpi_K$. Moreover, for $n \geq 4K^2$ every prime dividing $\Phi_n^{(K)}$ is removable, so D_n divides $D_n^{(K)}$.*
- (b) *$\lim_{n \rightarrow \infty} \frac{1}{n} \log D_n = 4 - \varpi$ with $\varpi = \lim_{K \rightarrow \infty} \varpi_K$. This is Theorem 1.4(e).*
- (c) *$\varpi = 0.29395989921603919\dots$, and $\varpi_K \geq 0.293957$ for $K = 10^5$.*

Sketch of proof; details in Appendix A.8. (a) For fixed k , the primes of a window contribute $\theta(\frac{3n}{3k+2}) - \theta(\frac{4n}{4k+3}) = \frac{n}{(3k+2)(4k+3)} + o(n)$ to $\log \Phi_n^{(K)}$, and similarly for the second kind; there are finitely many windows, and $\psi(4n) = 4n + o(n)$. A prime in a window with $k < K$ is larger than n/K , hence larger than $\sqrt{4n}$ when $n \geq 4K^2$, and Lemma 8.1 applies. (b) For $n \geq 4K^2$ the primes dividing Φ_n but not $\Phi_n^{(K)}$ are smaller than n/K , so $\log \Phi_n^{(K)} \leq \log \Phi_n \leq \log \Phi_n^{(K)} + \theta(n/K)$; divide by n and let first n , then K tend to infinity. (c) Numerical evaluation of the two series. $\square$

The formal proof uses only part (a) and the numerical bound in (c), with $K = 10^5$: it replaces D_n by the slightly larger multiplier $D_n^{(K)}$, for which no estimate of the tail of the series is needed. The price is $\varpi - \varpi_K < 1.4 \cdot 10^{-6}$ in the rate. Numerically, $\frac{1}{n} \log D_n = 3.70706, 3.70527, 3.70614, 3.70641$ for $n = 10^4, 10^5, 10^6, 10^7$ (limit 3.70604).

9. PROOF OF THEOREM 1.5

Let $K = 10^5$. For $n \geq 4K^2$ put

$$q_n := D_n^{(K)} v_n, \quad p_n := -D_n^{(K)} u_n.$$

These are integers: $D_n u_n$ and $D_n v_n$ are integers by Theorem 5.21, and D_n divides $D_n^{(K)}$ by Proposition 8.3(a). (For $n < 4K^2$ put $q_n = p_n = 1$; finitely many terms do not matter in Hata's lemma.) By Theorem 1.4(a),

$$q_n \pi - p_n = D_n^{(K)} (u_n + v_n \pi) = D_n^{(K)} J_n.$$

Growth. By Propositions 6.3 and 8.3(a),(c),

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log q_n = r + (4 - \varpi_K) =: \sigma, \quad 0 < \sigma \leq 10.540357 + (4 - 0.293957) = 14.2464.$$

(Here $\sigma > 0$ because $r > 0$ and $4 - \varpi_K = \lim \frac{1}{n} \log D_n^{(K)} \geq 0$, the numbers $D_n^{(K)}$ being positive integers.)

Decay. By Propositions 7.3 and 8.3(a), $|q_n \pi - p_n| \leq 4\rho^n D_n^{(K)}$, hence

$$\limsup_{n \rightarrow \infty} \frac{1}{n} \log |q_n \pi - p_n| \leq \log \rho + (4 - \varpi_K) =: -\tau, \quad \tau \geq 6.530134 - 3.706043 = 2.824091.$$

Conclusion. π is irrational (Lambert, 1761; for a short proof see [7]), so Lemma 1.2 gives

$$\mu(\pi) \leq 1 + \frac{\sigma}{\tau} \leq 1 + \frac{14.2464}{2.824091} = 6.04459665 \dots < 6.0446$$

(the arithmetic is checked in Appendix A.9). $\square$

One can also work with D_n itself and Proposition 8.3(b); then $\sigma \leq 14.246398$ and $\tau \geq 2.824093$, with the same conclusion.

Example 9.1 (the first integer forms). With the multiplier D_n the integer linear forms $D_n J_n = q_n \pi - p_n$ for $n = 1, 2, 3$ are

$$\begin{aligned} 44070 \pi - 138450 &= -0.01175 \dots, \\ 27617774940 \pi - 86763798860 &= 0.000292 \dots, \\ 84532652762578020 \pi - 265567160907372048 &= -0.0000339 \dots \end{aligned}$$

The first one is $390(113\pi - 355)$. The coefficients grow like $e^{14.25n}$ and the values decrease like $e^{-2.82n}$ (compare Table 7 for larger n).

Remark 9.2 (the exact rates). The three constants used in the proof are rounded values of

$$r = \log \lambda_3 = 10.5403551 \dots, \quad s = -\log |f(z_1)| = 6.5301361 \dots, \quad C = 4 - \varpi = 3.7060401 \dots$$

(Remarks 6.4 and 7.4, Proposition 8.3). Of these, only the value of C is proved in this paper (Proposition 8.3(b)). The values are related by $s = \frac{1}{2}(r + 20 \log 2 - 3 \log 3 - 5 \log 5)$, because the product of the three critical values of f is $3^3 5^5 / 2^{20}$. With these values the bound of Hata's lemma is

$$1 + \frac{r + C}{s - C} = 1 + \frac{14.2463952124 \dots}{2.8240960465 \dots} = 6.04458594096 \dots$$

To prove this slightly sharper bound one needs, instead of Proposition 7.1, the statement that on the polygon through the exact saddle point z_1 the maximum of $|f|$ is attained at z_1 . The original draft contains a numerical certificate that $\log |f| \leq -s + 5.4 \cdot 10^{-9}$ on that polygon (floating-point values of $\log |f|$ on a fine grid, with interval bounds for the derivative in between; this is not rigorous throughout). Its audit contains a certificate, computed with ball arithmetic (floating-point computation with rigorous error bounds), that the maximum is attained at z_1 . Neither is part of the formal development, and Theorem 1.5 claims only the rounded value. For orientation: without the removable primes ($C = 4$) the same formula gives 6.7469, and with the removable primes but without moving the path ($s = 6.0709$) it gives 7.0241.

10. FORMAL VERIFICATION

The proof of Theorem 1.5 has been formalised in the proof assistant Lean 4 (version 4.30.0) with the library Mathlib (release tag v4.30.0). This section says precisely what has been checked by the machine and what has not.

10.1. The formal statement. The final theorem of the development (file K2H/Main.lean, namespace PiMeasure) reads

```
theorem pi_irrationality_exponent
  (hPNT : Tendsto (fun x : ℝ => Chebyshev.theta x / x) atTop (ℕ 1)) :
  ∀ μ : ℝ, 6.0446 < μ → ExponentLE Real.pi μ
```

where

```
def ExponentLE (θ μ : ℝ) : Prop :=
  ∃ b₀ : ℕ, ∀ b : ℕ, b₀ ≤ b →
    ∀ a : ℤ, (b : ℝ) ^ (-μ) < |θ - a / b|
```

In words: *if $\theta(x)/x \rightarrow 1$ as $x \rightarrow \infty$, then for every real $\mu > 6.0446$ there is b_0 such that $|\pi - a/b| > b^{-\mu}$ for all integers a and all integers $b \geq b_0$.* By Definition 1.1 the conclusion is exactly $\mu(\pi) \leq 6.0446$. A corollary in the same file, `pi_not_liouvilleWith`, restates it with Mathlib’s predicate `LiouvilleWith`: under the same hypothesis π is not a Liouville number of exponent p for any $p > 6.0446$.

- **The only hypothesis is the prime number theorem**, in the form $\theta(x) \sim x$ for Chebyshev’s function (`Chebyshev.theta` of Mathlib). The prime number theorem is a classical theorem (Hadamard and de la Vallée Poussin, 1896), but it is not yet part of Mathlib; this is why it appears as a hypothesis and not as a lemma. It enters only through Proposition 8.3(a).
- Everything else is proved in the development: there is no `sorry` (Lean’s placeholder for a missing proof), and the command `#print axioms`, which lists the axioms on which a theorem depends, returns only `propext`, `Classical.choice` and `Quot.sound`. These are the three standard axioms of Lean’s logic, on which the library Mathlib as a whole is built.
- The development consists of the file `HataLemma.lean` and the 18 files of the directory K2H/, about 5,900 lines in total, together with a file `K2H.lean` of about 300 lines that contains the symmetry identity of Lemma 3.1 (for general exponents), the link to Mathlib’s `LiouvilleWith`, and earlier conditional versions of the last step. One of the files, `ContourCert.lean`, is generated by a script (it contains the 3×33 Bernstein coefficients of Proposition 7.1).
- The whole library compiles with Lean’s build tool (`lake build`), both inside the working repository and as a standalone project (directory `lean/` of the accompanying material, Section 1.7, with a recorded build log); the file `README.md` there contains a map of the modules.

10.2. The modules. Table 6 lists the files, their main statements and the corresponding statements of this paper. The proofs in Sections 4–9 and in Appendix A were written to follow the formal proofs step by step. The definitions of the formal objects agree with those of this paper: `f534` is f , `J534 n` is J_n (defined as the integral $-i \int_{-1}^1 f(1+ti)^n(1+ti)^{-1} i dt$), `RemovablePrime`, `Phi` and `Dmult` are Definition 1.3, and `cPS n` is the power-series coefficient of Lemma 6.1.

file	main statements (informally)	in this paper
<code>HataLemma.lean</code>	<code>exponent_le_of_forms</code> : Hata’s lemma in ε – N form	Lemma 1.2, App. A.9
<code>Defs.lean</code>	definitions of f , J_n , removable primes, Φ_n , D_n , c_n	Section 1
<code>PInt.lean</code>	p -integers of $\mathbb{C}$; the levels $\mathfrak{L}_k$; a rational number that is p -integral for all p is an integer	Lemmas 5.2, 5.17
<code>LinearForm.lean</code>	<code>primPair_unique</code> ; <code>J534.eq_of_primPair</code> : $J_n = -i(G(1+i) - G(1-i)) + c\pi/2$	Lemmas 4.2, 4.3
<code>QForm.lean</code>	<code>exists_primitive_WY</code> (the recursion over a subring R); <code>exists_rational_primPair</code>	Prop. 4.6, Cor. 4.7
<code>PF.lean</code>	partial fractions on Ω : existence over a subring, uniqueness, derivative, standard primitive	Lemma 5.5
<code>OddPrimes.lean</code>	<code>oddPrime_trivial</code> ; <code>removable_decomp</code> (the identity (5.6) for the decompositions); <code>oddPrime_removable</code>	Props. 5.7, 5.9
<code>TwoAdic.lean</code>	<code>twoAdic_primitive</code> : a primitive pair with c and $2^{\lfloor \log_2 4n \rfloor} u_n$ 2-integral	Prop. 5.12
<code>Integrality.lean</code>	<code>theoremA</code> : $J_n = u + \frac{c}{2}\pi$ with $u, c \in \mathbb{Q}$, $D_n u \in \mathbb{Z}$, $D_n \frac{c}{2} \in \mathbb{Z}$	Theorem 5.21
<code>ResidueLink.lean</code> , <code>Forms.lean</code>	<code>residue_eq_cPS</code> : the constant of a primitive pair is the power-series coefficient; <code>linear_forms</code>	Lemma 6.1
<code>Growth.lean</code>	<code>cPS_pos</code> , <code>cPS_supermul</code> , <code>cPS_growth</code> : the limit r exists and $0 < r \leq 10.540357$	Prop. 6.3
<code>ContourDefs.lean</code> , <code>ContourCert.lean</code> , <code>ContourBound.lean</code>	<code>f534.le_seg1</code> , <code>f534.le_seg2</code> : $ f \leq \rho$ on the two segments; <code>rho.le</code> : $\rho \leq e^{-6.530134}$	Prop. 7.1
<code>Decay.lean</code>	<code>J534_norm.le</code> : $ J_n \leq 4\rho^n$	Prop. 7.3
<code>MultRate.lean</code>	<code>windowPrime_removable</code> ; <code>DmultK.rate</code> : $\frac{1}{n} \log D_n^{(K)} \rightarrow 4 - \varpi_K$ from the hypothesis $\theta(x) \sim x$; <code>varpiK.lower</code> : $\varpi_{10^5} \geq 0.293957$	Prop. 8.3(a),(c)
<code>Reduction.lean</code> , <code>Main.lean</code>	<code>exponent_of_linear_forms</code> ; <code>pi_irrationality_exponent</code> ; <code>pi_not_liouvilleWith</code>	Section 9

TABLE 6. The Lean development (directory `lean/MuPi` of the accompanying material, namespace `PiMeasure`; all files except `HataLemma.lean` lie in the sub-directory `K2H/`).

A few points where the formal proof and the text differ in inessential ways: in Lean the lower bound for c_1 is 16 instead of the exact value 7345; the inequality $c_n \leq f(-x)^n$ is proved through partial sums instead of convergent series; the vanishing of the constant c_ω in the proof of Proposition 5.9 and the equality $b_1 = b'_1 = 0$ are obtained from the uniqueness of partial fractions, as here; and the numerical inequalities $37811 \leq e^{10.540357}$ and $\rho \leq e^{-6.530134}$ are derived from Mathlib’s bounds $2.7182818283 < e < 2.7182818286$ and Taylor polynomials (Appendix A.7).

10.3. What is not covered.

- *The prime number theorem* (hypothesis, see above).
- *The sharper statements* that are mentioned but not needed for Theorem 1.5: Proposition 8.3(b), that is Theorem 1.4(e) (the rate of the full multiplier D_n ; it is proved in Appendix A.8, but the formal proof works with the truncated multiplier $D_n^{(K)}$ of Proposition 8.3(a)); the exact values of r and s (Remarks 6.4, 7.4, 9.2); and all of Appendices B–D.
- *This text.* The machine has checked the Lean files, not the present exposition. We have tried to make the two agree, and all numbers printed here have been recomputed (Section 11), but misprints in the text are possible.
- *Human judgement.* A formal proof guarantees that the stated theorem follows from the stated hypothesis and the axioms of Lean. It does not replace the reading of an expert, who would also examine whether the definitions say what they are meant to say, how the result relates to the literature, and whether it is new. That reading has not yet taken place. The whole work (search, proofs, formalisation and text) was generated by AI systems under human direction.

11. NUMERICAL CHECKS

All numbers in this paper were recomputed for this version, with exact rational arithmetic wherever possible (short Python programs using exact fractions, the library FLINT for polynomial arithmetic, and multiprecision floating point for the constants).

n	$\frac{1}{n} \log v_n$	$-\frac{1}{n} \log J_n $	$\frac{1}{n} \log D_n$	$\frac{-\log D_n J_n }{\log(D_n v_n)}$	sign of J_n
1	8.2086	6.9283	2.4849	0.4155	—
5	9.9145	6.7855	3.4639	0.2483	—
10	10.1928	6.7369	3.6215	0.2255	+
20	10.3493	6.7032	3.8049	0.2048	—
50	10.4548	6.5687	3.7261	0.2004	—
100	10.4941	6.5537	3.6743	0.2032	+
200	10.5155	6.5628	3.7254	0.1992	+
400	10.5271	6.5382	3.7177	0.1980	—
predicted	10.5404	6.5301	3.7060	0.1982	

TABLE 7. Exact data against the rates predicted in Remark 9.2 (last row). The slow convergence of $\frac{1}{n} \log v_n$ is explained by the factor $n^{-1/2}$ in Remark 6.4; $\frac{1}{n} \log D_n$ fluctuates because of the irregular distribution of the primes; the entry of $-\frac{1}{n} \log |J_n|$ for $n = 200$ is larger than its neighbours because the pattern of signs changes there (J_{199} and J_{200} have the same sign, compare Appendix B). The column before the last is the ratio of the decay rate to the growth rate of the integer forms $D_n J_n$; its predicted value $(s - C)/(r + C) = 0.19823$ corresponds to the exponent $1 + 1/0.19823 = 6.0446$. What is proved in this paper is less than the last row: $\frac{1}{n} \log v_n$ has a limit $r \leq 10.540357$; $-\frac{1}{n} \log |J_n| \geq 6.530134 - \frac{1}{n} \log 4$ for all n ; and $\frac{1}{n} \log D_n \rightarrow 3.7060 \dots$ (assuming the prime number theorem).

- *The forms.* The pairs (u_n, c_n) were computed by three independent methods: the recursion of Section 4.2 (all $n \leq 30$); the closed formula of Remark 4.9 (all $n \leq 300$); and the

decomposition in the coordinate w of Section 5.4, in exact arithmetic in $\mathbb{Q}(\sqrt{2})$ ($n \leq 6$). For $n = 1, 2$ also the partial fractions (5.3) were computed in $\mathbb{Q}(\sqrt{2})$. All results agree, and for $n = 1, 2, 3$ they agree with a numerical quadrature of the integral (1.2) to 38 decimal places.

- *Integrality.* $D_n u_n$ and $D_n v_n$ are integers for every $n \leq 300$; no removable prime divides the denominator of u_n ; $2^{\lfloor \log_2(4n) \rfloor} u_n$ has odd denominator and c_n is an integer in all these cases. The levels asserted in Lemma 5.18 and in (5.13) hold for $n \leq 6$; the level $14n$ of V is attained for $n = 1, 2, 3, 4, 6$ (for $n = 5$ the integer c_5 is even and the level is $14n + 2$).
- *The certificate.* The Bernstein coefficients of Proposition 7.1 were recomputed with a separate program; they coincide with the 99 rational numbers in the Lean file, and all are positive.
- *Sizes.* Table 7 shows the three rates for some values of n , computed from the exact forms (the values of J_n were evaluated with sufficiently many digits to control the cancellation between u_n and $v_n \pi$).

All checks of this section, and the other numbers printed in this paper including those of the appendices, are recomputed by the program `scripts/paper_v3_numbers.py` of the accompanying material (Section 1.7). It was written for an audit of the previous version of this text, independently of the programs that first produced the numbers, and runs in less than a minute.

The original draft of this work was examined by two independent AI audits, which re-derived the lemmas of the classical version of the proof and report exact verifications of Theorem 1.4(b) for all $n \leq 800$ and for selected n up to 5000. We have not repeated these larger computations for the present version.

APPENDIX A. DETAILED DERIVATIONS FOR NON-EXPERTS

This appendix contains, line by line, the computations that the main text calls “direct” or “standard”. It is meant for readers who want to check every step by hand; nothing here is new. The subsections follow the order of the main text.

A.1. The algebra of W and Y ; the case $n = 1$.

Proof of Lemma 3.2. (a) $W^2 - Y^2 = (W - Y)(W + Y) = \frac{4}{z} \cdot 2z = 8$.

(b) $\frac{dW}{dz} = 1 - \frac{2}{z^2}$, so $z \frac{dW}{dz} = z - \frac{2}{z} = Y$; and $\frac{dY}{dz} = 1 + \frac{2}{z^2}$, so $z \frac{dY}{dz} = z + \frac{2}{z} = W$.

(c) $W - 3 = \frac{z^2 - 3z + 2}{z} = \frac{(z-1)(z-2)}{z}$, $W - 2 = \frac{z^2 - 2z + 2}{z}$, $Y = \frac{z^2 - 2}{z}$. Therefore

$$\frac{(W - 3)^5 (W - 2)^3}{Y^4} = \frac{((z-1)(z-2))^5}{z^5} \cdot \frac{(z^2 - 2z + 2)^3}{z^3} \cdot \frac{z^4}{(z^2 - 2)^4} = \frac{N(z)}{z^4(z^2 - 2)^4} = f(z).$$

(d) $\frac{2}{1+i} = \frac{2(1-i)}{(1+i)(1-i)} = 1 - i$. Hence $W(1+i) = (1+i) + (1-i) = 2$ and $Y(1+i) = (1+i) - (1-i) = 2i$; exchanging the roles, $W(1-i) = 2$ and $Y(1-i) = -2i$.

The derivative identity (4.3). By the product rule and (b),

$$\begin{aligned} \frac{d}{dz} (W^{j+1} Y^{-2m-1}) &= (j+1) W^j \cdot \frac{Y}{z} \cdot Y^{-2m-1} - (2m+1) W^{j+1} Y^{-2m-2} \cdot \frac{W}{z} \\ &= (j+1) \frac{W^j}{Y^{2m} z} - (2m+1) \frac{W^{j+2}}{Y^{2m+2} z}. \end{aligned}$$

By (a), $W^{j+2} = W^j(Y^2 + 8)$, so the last fraction is $\frac{W^j}{Y^{2m}z} + 8\frac{W^j}{Y^{2m+2}z} = X_j^{(m)} + 8X_j^{(m+1)}$. Altogether

$$\begin{aligned} \frac{d}{dz}(W^{j+1}Y^{-2m-1}) &= (j+1)X_j^{(m)} - (2m+1)X_j^{(m)} - 8(2m+1)X_j^{(m+1)} \\ &= (j-2m)X_j^{(m)} - 8(2m+1)X_j^{(m+1)}. \end{aligned}$$

The base case (4.4). $W^j = (z + 2z^{-1})^j = \sum_k \binom{j}{k} z^{j-k} (2z^{-1})^k = \sum_k \binom{j}{k} 2^k z^{j-2k}$; divide by z . For example $X_2^{(0)} = W^2/z = z + 4z^{-1} + 4z^{-3}$: the first and the third term are the derivatives of $z^2/2$ and $-2z^{-2}$, and the middle term, which has no rational primitive, gives the constant $c = 4 = \binom{2}{1} \cdot 2$.

The case $n = 1$ in full. Here $h_1 = P(W)Y^{-4}/z$ with $P = P_1 = \sum_{j=0}^8 p_j W^j$,

$$(p_0, \dots, p_8) = (1944, -6156, 8478, -6633, 3225, -998, 192, -21, 1).$$

Applying (4.5) with $m = 0$ and then with $m = 1$:

$$\begin{aligned} X_j^{(1)} &= \frac{1}{8} [j X_j^{(0)} - (W^{j+1}Y^{-1})'], \\ X_j^{(2)} &= \frac{1}{24} [(j-2) X_j^{(1)} - (W^{j+1}Y^{-3})'] = \frac{j(j-2)}{192} X_j^{(0)} - \frac{j-2}{192} (W^{j+1}Y^{-1})' - \frac{1}{24} (W^{j+1}Y^{-3})'. \end{aligned}$$

Multiply by p_j and add:

$$(A.1) \quad h_1 = \frac{A(W)}{z} - \left(\frac{B(W)}{Y} + \frac{C(W)}{Y^3} \right)',$$

where

$$\begin{aligned} A(W) &= \sum_j \frac{j(j-2)}{192} p_j W^j, & C(W) &= \frac{1}{24} W P(W), \\ B(W) &= \sum_j \frac{j-2}{192} p_j W^{j+1} = \frac{1}{192} (W^2 P'(W) - 2W P(W)). \end{aligned}$$

With the numbers p_j ,

$$A(W) = \frac{1}{4} W^8 - \frac{245}{64} W^7 + 24W^6 - \frac{2495}{32} W^5 + \frac{1075}{8} W^4 - \frac{6633}{64} W^3 + \frac{513}{16} W.$$

The bracket vanishes at the end points. $P = (W-3)^5(W-2)^3$ has a triple zero at $W = 2$, so $P(2) = P'(2) = 0$ and hence $B(2) = C(2) = 0$. Since $W(1 \pm i) = 2$ and $Y(1 \pm i) = \pm 2i \neq 0$, the function $B(W)/Y + C(W)/Y^3$ is zero at $z = 1 + i$ and at $z = 1 - i$.

The Laurent polynomial. Expanding each power of W by (4.4) gives $A(W)/z = \sum_{l=-8}^8 \Lambda_l z^{l-1}$ with

l	0	1	2	3	4	5	6	7	8		
Λ_l	7345	$-\frac{152973}{32}$	2627	$-\frac{77113}{64}$	$\frac{3603}{8}$	$-\frac{2105}{16}$	28	$-\frac{245}{64}$	$\frac{1}{4}$	and	$\Lambda_{-l} = 2^l \Lambda_l$.

Explicitly, if $A(W) = \sum_j a_j W^j$, then by (4.4) $\Lambda_l = \sum_j a_j \binom{j}{k} 2^k$, where the sum is over the $j \geq |l|$ with $j-l$ even, and $k = (j-l)/2$. For instance $\Lambda_8 = \frac{1}{4}$, $\Lambda_6 = 24 + \frac{1}{4} \binom{8}{1} \cdot 2 = 28$,

$$\Lambda_1 = \frac{513}{16} - \frac{6633}{64} \binom{3}{1} 2 - \frac{2495}{32} \binom{5}{2} 2^2 - \frac{245}{64} \binom{7}{3} 2^3 = \frac{2052-39798-199600-68600}{64} = -\frac{152973}{32},$$

and $\Lambda_0 = \frac{1075}{8} \binom{4}{2} 2^2 + 24 \binom{6}{3} 2^3 + \frac{1}{4} \binom{8}{4} 2^4 = 3225 + 3840 + 280 = 7345$. The relation $\Lambda_{-l} = 2^l \Lambda_l$ holds because $A(W) = \sum_l \Lambda_l z^l$ does not change when z is replaced by $2/z$: the replacement turns the sum into $\sum_l \Lambda_l 2^l z^{-l}$, and comparing the coefficients of z^{-l} gives $\Lambda_{-l} = 2^l \Lambda_l$.

The primitive pair. By (A.1), $h_1 = 7345/z + G'$ with

$$G(z) = \sum_{l \neq 0} \frac{\Lambda_l}{l} z^l - \frac{B(W)}{Y} - \frac{C(W)}{Y^3}.$$

So $(G, 7345)$ is a primitive pair and $c_1 = 7345$, $v_1 = \frac{7345}{2}$.

The value of u_1 . We have $(1+i)^l - (1-i)^l = 2i \operatorname{Im}(1+i)^l$, and $\operatorname{Im}(1+i)^l = 1, 2, 2, 0, -4, -8, -8, 0$ for $l = 1, \dots, 8$. For negative exponents, $(1 \pm i)^{-l} = (1 \mp i)^l/2^l$, so $(1+i)^{-l} - (1-i)^{-l} = -2i \operatorname{Im}(1+i)^l/2^l$; together with $\Lambda_{-l} = 2^l \Lambda_l$ this shows that the terms l and $-l$ of G contribute equally to $G(1+i) - G(1-i)$. Therefore

$$\begin{aligned} u_1 &= -i(G(1+i) - G(1-i)) = 4 \sum_{l=1}^8 \frac{\Lambda_l}{l} \operatorname{Im}(1+i)^l \\ &= 4 \left(-\frac{152973}{32} + 2627 - \frac{77113}{96} + \frac{421}{4} - \frac{112}{3} + \frac{35}{8} \right) = 4 \cdot \left(-\frac{23075}{8} \right) = -\frac{23075}{2}. \end{aligned}$$

(The six terms come from $l = 1, 2, 3, 5, 6, 7$; over the common denominator 96 the numerators are $-458919 + 252192 - 77113 + 10104 - 3584 + 420 = -276900$.) By Lemma 4.3,

$$J_1 = -\frac{23075}{2} + \frac{7345}{2} \pi = \frac{7345\pi - 23075}{2}.$$

General n : Remark 4.9. Iterating (4.5) m times gives, by induction on m ,

$$X_j^{(m)} = \xi_{j,m} X_j^{(0)} - \sum_{m'=0}^{m-1} \zeta_{j,m,m'} \left(\frac{W^{j+1}}{Y^{2m'+1}} \right)', \quad \zeta_{j,m,m'} = \frac{1}{8(2m'+1)} \prod_{i=m'+1}^{m-1} \frac{j-2i}{8(2i+1)},$$

and $\zeta_{j,m,m'}$ is a polynomial in j of degree $m-1-m' \leq m-1$. Hence $h_n = A_n(W)/z - E'_n$ with $E_n = \sum_{m' < 2n} B_{m'}(W) Y^{-(2m'+1)}$ and $B_{m'}(W) = \sum_j \zeta_{j,2n,m'} p_j W^{j+1}$. For a polynomial $q(j)$ of degree e , the sum $\sum_j q(j) p_j W^j$ is a combination of $W^i P_n^{(i)}(W)$ with $i \leq e$: indeed $\sum_j j(j-1) \cdots (j-i+1) p_j W^j = W^i P_n^{(i)}(W)$, and every polynomial $q(j)$ of degree e is a combination of the products $j(j-1) \cdots (j-i+1)$ with $i \leq e$. As P_n has a zero of order $3n$ at $W = 2$ and $e \leq 2n-1 < 3n$, all $B_{m'}$ vanish at $W = 2$, that is, $E_n(1 \pm i) = 0$. Thus $J_n = -i \int_{1-i}^{1+i} A_n(W) dz/z$. The function $A_n(W)/z$ is a Laurent polynomial in z ; on $\operatorname{Re} z > 0$ it has the primitive $c_n \operatorname{Log} z + \sum_{l \neq 0} (\Lambda_l/l) z^l$, so the integral may be taken along the arc $z = \sqrt{2} e^{i\varphi}$, $-\frac{\pi}{4} \leq \varphi \leq \frac{\pi}{4}$, on which $dz/z = i d\varphi$ and $W = 2\sqrt{2} \cos \varphi$. Finally, for $j = 2l$,

$$\xi_{2l,m} = \prod_{i=0}^{m-1} \frac{2(l-i)}{8(2i+1)} = \frac{1}{4^m} \cdot \frac{l!}{(l-m)!} \cdot \frac{2^m m!}{(2m)!} = 2^{-m} \binom{l}{m} / \binom{2m}{m} \quad (l \geq m),$$

and $\xi_{2l,m} = 0$ for $l < m$; inserting this in (4.6) with $m = 2n$ gives the formula for c_n in Remark 4.9.

A.2. Primitive pairs.

The circle integral in Lemma 4.2. Let $\Delta = G_1 - G_2$, so $\Delta'(z) = (c_2 - c_1)/z$ on Ω . The circle $z(t) = \frac{1}{2} e^{it}$, $0 \leq t \leq 2\pi$, avoids 0 and $\pm\sqrt{2}$ (because $\frac{1}{2} < \sqrt{2}$). By the chain rule $\frac{d}{dt} \Delta(z(t)) = \Delta'(z(t)) z'(t)$, hence

$$0 = \Delta(z(2\pi)) - \Delta(z(0)) = \int_0^{2\pi} \Delta'(z(t)) z'(t) dt = (c_2 - c_1) \int_0^{2\pi} \frac{\frac{i}{2} e^{it}}{\frac{1}{2} e^{it}} dt = 2\pi i (c_2 - c_1).$$

So $c_1 = c_2$.

The strip. Now $\Delta' = 0$ on Ω . Let $S = \{z : \frac{1}{2} < \operatorname{Re} z < \frac{5}{4}\}$. It does not contain 0 (real part 0), $-\sqrt{2}$ (negative real part) or $\sqrt{2}$ (since $\sqrt{2} = 1.414 \dots > \frac{5}{4}$), so $S \subset \Omega$; and $1 \pm i \in S$. S is convex, so the segment from $1 - i$ to $1 + i$ lies in S and $\Delta(1 + i) - \Delta(1 - i) = \int_0^1 \Delta'(1 - i + 2it) \cdot 2i dt = 0$. (We avoid in this way the statement that Ω is connected.)

The logarithms. $1 \pm i = \sqrt{2} e^{\pm i\pi/4}$, so the principal logarithm is $\operatorname{Log}(1 \pm i) = \log \sqrt{2} \pm i\frac{\pi}{4}$, and $\operatorname{Log}(1 + i) - \operatorname{Log}(1 - i) = i\frac{\pi}{2}$. On the half-plane $\operatorname{Re} z > 0$ the function Log is holomorphic with $\operatorname{Log}'(z) = 1/z$. Hence $-i \cdot c \cdot i\frac{\pi}{2} = c\frac{\pi}{2}$ in Lemma 4.3.

A.3. p -integers and partial fractions.

Proof of Lemma 5.2. (a) If Mx and $M'x'$ are algebraic integers with $p \nmid M$, $p \nmid M'$, then $MM'(x \pm x') = M'(Mx) \pm M(M'x')$ and $MM'xx' = (Mx)(M'x')$ are algebraic integers, and $p \nmid MM'$. The numbers $\sqrt{2}$ and i are roots of $X^2 - 2$ and $X^2 + 1$; and $M \cdot \frac{1}{M} = 1$. (b) If x is rational and Mx is an algebraic integer, then Mx is an integer k , and $x = k/M$ has a denominator dividing M , hence prime to p . Conversely, if $x = a/b$ with $p \nmid b$, then $bx = a$. (c) A positive integer divisible by no prime is 1.

What this means in practice. For odd p the numbers $\frac{1}{2}$, $\frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$, $\frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$ and $\frac{1}{1 \pm i} = \frac{1 \mp i}{2}$ are p -integral: their only denominators are powers of 2. They are not 2-integral; for instance, if $M/\sqrt{2}$ were an algebraic integer for an odd M , so would be its square $M^2/2$, which is rational but not an integer. This is the precise sense in which 2 is the only bad prime for the configuration $\{0, \pm\sqrt{2}, 1 \pm i\}$. The identities (5.4) follow from

$$(1 \pm i - \sqrt{2})(1 \pm i + \sqrt{2}) = (1 \pm i)^2 - 2 = -2(1 \mp i), \quad (1 \mp i)(1 \pm i) = 2,$$

which give $(1 \pm i - \sqrt{2})(1 \pm i + \sqrt{2})(1 \pm i) = -4$.

Proof of Lemma 5.5. Existence, with coefficients in R . Let $\varphi = M(z)/\prod_b(z - b)^{e_b}$, the product over $b \in \{0, \sqrt{2}, -\sqrt{2}\}$. Fix a pole a and put $t = z - a$. For $b \neq a$ the difference $a - b$ is one of $\pm\sqrt{2}$, $\pm 2\sqrt{2}$, and its inverse lies in R because $\sqrt{2}, \frac{1}{2} \in R$. Hence

$$\frac{1}{z - b} = \frac{1}{(a - b) + t} = \frac{1}{a - b} \sum_{k \geq 0} \left(\frac{-t}{a - b} \right)^k$$

is a power series in t with coefficients in R , and so is $M(a + t) \prod_{b \neq a} (z - b)^{-e_b} = \sum_{k \geq 0} s_k t^k$. Thus $\varphi = t^{-e_a} \sum_k s_k t^k$ near a , and we define the principal part at a by $\sum_{k=1}^{e_a} q_{a,k} (z - a)^{-k}$ with $q_{a,k} := s_{e_a - k} \in R$ (an empty sum if $e_a = 0$). The difference of φ and its three principal parts is a rational function without poles, that is a polynomial $T(z)$. To see that T has coefficients in R , multiply by the monic polynomial $D(z) = \prod_b (z - b)^{e_b} \in R[z]$: the product $T \cdot D = M - \sum_a (\text{principal part at } a) \cdot D$ is a polynomial with coefficients in R , and the quotient of a polynomial in $R[z]$ by a monic polynomial in $R[z]$ has coefficients in R .

Uniqueness. Suppose a function of the form (5.1) vanishes identically on Ω . If some $q_{a,k} \neq 0$, let k be maximal for this a ; as $z \rightarrow a$ the term $q_{a,k}(z - a)^{-k}$ tends to infinity faster than all other terms, which is impossible. So all $q_{a,k} = 0$, and a polynomial with infinitely many zeros is 0.

(i) $\frac{d}{dz}(z - a)^{-k} = -k(z - a)^{-k-1}$: differentiation raises the order of every polar term by one, so no term of order 1 arises, and the result is again of the form (5.1). (ii) Differentiate (5.2) term by term: $\frac{d}{dz} \frac{z^{j+1}}{j+1} = z^j$ and $\frac{d}{dz} \left(-\frac{1}{(k-1)(z-a)^{k-1}} \right) = \frac{1}{(z-a)^k}$ for $k \geq 2$. All terms of (5.1) are recovered except the terms with $k = 1$.

The integration indices of h_n . In (5.3): the polynomial part has degree $4n - 1$, so the indices $j + 1$ run through $1, \dots, 4n$; the pole at 0 has order $4n + 1$, so the indices $k - 1$ run through $1, \dots, 4n$; the poles at $\pm\sqrt{2}$ have order $4n$, so $k - 1$ runs through $1, \dots, 4n - 1$. No index exceeds $4n$. For $n = 1$ (Example 5.8) the primitive is

$$G_{h_1}(z) = \frac{z^4}{4} - 7z^3 + 108z^2 - 1460z + \frac{2920}{z} - \frac{432}{z^2} + \frac{56}{z^3} - \frac{4}{z^4} - \sum_{k=2}^4 \frac{1}{k-1} \left(\frac{b_k}{(z - \sqrt{2})^{k-1}} + \frac{\bar{b}_k}{(z + \sqrt{2})^{k-1}} \right),$$

and evaluating $-i(G_{h_1}(1+i) - G_{h_1}(1-i))$ in exact arithmetic gives again $-\frac{23075}{2}$.

A.4. Removable primes. We comment on the four steps of the proof of Proposition 5.9 and then work out an example.

Step 1. $h_n = (W-3)^{5n}(W-2)^{3n}Y^{-4n}/z$, and $5n = pq_1 + \alpha_1$ gives $(W-3)^{5n} = ((W-3)^{q_1})^p(W-3)^{\alpha_1}$; similarly for the other two factors. In terms of z ,

$$g = \frac{((z-1)(z-2))^{q_1}(z^2-2z+2)^{q_2}z^{q_0-q_1-q_2}}{(z^2-2)^{q_0}}$$

is a rational function with integer coefficients and poles only at $0, \pm\sqrt{2}$ (the exponent $q_0 - q_1 - q_2$ may be negative).

Step 2. Using Lemma 3.2(c), $\omega = ((z-1)(z-2))^{\alpha_1}z^{-\alpha_1} \cdot (z^2-2z+2)^{\alpha_2}z^{-\alpha_2} \cdot z^\delta(z^2-2)^{-\delta} \cdot z^{-1}$, which is the formula in the text; the exponent of z is $\delta - \alpha_1 - \alpha_2 - 1 \geq 0$, so there is no pole at 0. (At infinity ω behaves like $z^{\alpha_1+\alpha_2-\delta-1}$ with exponent ≤ -2 , so ωdz has no pole at ∞ either; this explains why G_{red} remains bounded as $z \rightarrow \infty$, but it is not needed.) The recursion of Proposition 4.6 divides only by 8, by integers $l \leq \alpha_1 + \alpha_2 \leq p - 2$ and by odd integers $\leq \delta - 1 \leq p - 2$; so G_{red} is a combination with p -integral coefficients of powers of z and of the functions $W^{j+1}Y^{-(2m'+1)}$, all of which belong to $\mathcal{A}(R_p)$. The vanishing of the constant c_ω can also be read off from (4.6): an even $j \leq \alpha_1 + \alpha_2 \leq \delta - 1$ satisfies $j \leq \delta - 2 = 2m - 2$, so the product $\xi_{j,m}$ contains the factor $j - 2i = 0$ for $i = j/2$.

Steps 3 and 4. $H' = (g^p)'G_{\text{red}} + g^pG'_{\text{red}} = p g^{p-1}g'G_{\text{red}} + g^p\omega = p\eta + h_n$. For the comparison of coefficients, differentiate the decomposition (5.1) of H term by term:

$$H' = \sum_j (j+1) t_{j+1}^H z^j - \sum_a \sum_{k \geq 2} (k-1) q_{a,k-1}^H (z-a)^{-k}.$$

Subtracting p times the decomposition of η gives a decomposition of h_n , which by uniqueness is (5.3).

A complete example: $n = 2, p = 3$. The data are $q_1 = 3, \alpha_1 = 1, q_2 = 2, \alpha_2 = 0, q_0 = 2, \delta = 2$ (Example 5.10), so

$$g = \frac{((z-1)(z-2))^3(z^2-2z+2)^2}{z^3(z^2-2)^2}, \quad \omega = \frac{W-3}{Y^2} \cdot \frac{1}{z} = X_1^{(1)} - 3X_0^{(1)}.$$

By (4.5) with $m = 0$: $X_0^{(1)} = -\frac{1}{8}(W/Y)'$ and $X_1^{(1)} = \frac{1}{8}[X_1^{(0)} - (W^2/Y)']$, where $X_1^{(0)} = W/z = 1 + 2z^{-2} = Y'$. Hence

$$\omega = \frac{1}{8} \left(Y - \frac{W^2}{Y} + \frac{3W}{Y} \right)' = \left(\frac{3W-8}{8Y} \right)', \quad G_{\text{red}} = \frac{3W-8}{8Y} = \frac{3z^2-8z+6}{8(z^2-2)},$$

using $Y^2 - W^2 = -8$. One checks directly that $G'_{\text{red}} = (z-1)(z-2)/(z^2-2)^2 = \omega$. The polynomial parts of h_2 , of $H = g^3 G_{\text{red}}$ and of $\eta = g^2 g' G_{\text{red}}$ are

$$\begin{aligned} T_h &= z^7 - 42z^6 + 873z^5 - 11992z^4 + 122666z^3 - 998316z^2 + 6745574z - 38995784, \\ T_H &= \frac{3}{8}z^9 - \frac{125}{8}z^8 + \frac{1287}{4}z^7 - \frac{8743}{2}z^6 + \frac{176593}{4}z^5 - \frac{1416151}{4}z^4 + 2352203z^3 - \frac{26681977}{2}z^2 + \dots, \\ T_\eta &= \frac{9}{8}z^8 - 42z^7 + \frac{3059}{4}z^6 - 9034z^5 + \frac{310311}{4}z^4 - 512939z^3 + 2684975z^2 - \dots, \end{aligned}$$

and indeed $T_h = T'_H - 3T_\eta$. Look at the two coefficients of T_h whose integration index is divisible by 3:

$$t_2 = 3 \cdot 2352203 - 3 \cdot 2684975 = -998316, \quad t_5 = 6 \cdot \left(-\frac{8743}{2}\right) - 3 \cdot (-9034) = 873.$$

After division by the index, as in (5.7): $\frac{t_2}{3} = 2352203 - 2684975 = -332772$ and $\frac{t_5}{6} = -\frac{8743}{2} + \frac{1}{2} \cdot 9034 = \frac{291}{2}$, both without a 3 in the denominator. The same happens at the pole 0, where the coefficients of h_2 are

$$(a_1, \dots, a_9) = (197269821, -77991568, 26982296, -7986528, 1962656, -383744, 55872, -5376, 256) :$$

the indices divisible by 3 belong to $a_4 = -3 \cdot 2662176$ and $a_7 = 3 \cdot 18624$, while for example a_2 and a_3 are not divisible by 3. (Also $a_1 = c_2 = 3 \cdot 65756607$, in accordance with Remark 5.11.) At the poles $\pm\sqrt{2}$ the coefficients with index 3 and 6 are $b_4 = -\frac{99124099977}{64} + \frac{280365293091}{256}\sqrt{2}$ and $b_7 = \frac{13435662249}{8} - \frac{4750223943}{4}\sqrt{2}$, whose numerators are all divisible by 3. The outcome is $u_2 = -\frac{4338189943}{14}$: no 3 in the denominator.

A.5. The prime 2: the coordinate w . Throughout, $\varepsilon = \sqrt{2} - 1$, so $\varepsilon^2 = 3 - 2\sqrt{2}$ and

$$(A.2) \quad \varepsilon^2 + 2\varepsilon = 1, \quad 1 + \varepsilon^2 = 4 - 2\sqrt{2} = 2\sqrt{2}\varepsilon = \varkappa, \quad 1 - \varepsilon^2 = 2\varepsilon.$$

Proof of Lemma 5.13. From (5.8), $w^2 = -\varepsilon^2(\sqrt{2} + z)^2/(\sqrt{2} - z)^2$. We compute the four factors of f in terms of w .

$$\begin{aligned} w^2 + \varepsilon^2 &= \varepsilon^2 \frac{(\sqrt{2} - z)^2 - (\sqrt{2} + z)^2}{(\sqrt{2} - z)^2} = -\frac{4\sqrt{2}\varepsilon^2 z}{(\sqrt{2} - z)^2}, \\ 1 - w^2 &= \frac{(\sqrt{2} - z)^2 + \varepsilon^2(\sqrt{2} + z)^2}{(\sqrt{2} - z)^2} = \frac{(1 + \varepsilon^2)(z^2 + 2) - 2\sqrt{2}(1 - \varepsilon^2)z}{(\sqrt{2} - z)^2} = \frac{2\sqrt{2}\varepsilon(z^2 - 2z + 2)}{(\sqrt{2} - z)^2}, \\ 1 + \varepsilon^2 w^2 &= \frac{(\sqrt{2} - z)^2 - \varepsilon^4(\sqrt{2} + z)^2}{(\sqrt{2} - z)^2} = \frac{(1 - \varepsilon^4)(z^2 + 2) - 2\sqrt{2}(1 + \varepsilon^4)z}{(\sqrt{2} - z)^2} \\ &= \frac{4\sqrt{2}\varepsilon^2(z-1)(z-2)}{(\sqrt{2} - z)^2}, \\ w^4 &= \frac{\varepsilon^4(\sqrt{2} + z)^4}{(\sqrt{2} - z)^4}. \end{aligned}$$

In the second line we used (A.2): $(1 + \varepsilon^2)(z^2 + 2) - 2\sqrt{2} \cdot 2\varepsilon z = 2\sqrt{2}\varepsilon(z^2 + 2 - 2z)$. In the third line, $1 - \varepsilon^4 = (1 - \varepsilon^2)(1 + \varepsilon^2) = 4\sqrt{2}\varepsilon^2$ and $1 + \varepsilon^4 = (1 + \varepsilon^2)^2 - 2\varepsilon^2 = 8\varepsilon^2 - 2\varepsilon^2 = 6\varepsilon^2$, so the numerator is $4\sqrt{2}\varepsilon^2(z^2 + 2) - 12\sqrt{2}\varepsilon^2 z = 4\sqrt{2}\varepsilon^2(z^2 - 3z + 2)$. The first line shows $w^2 + \varepsilon^2 \neq 0$ for $z \neq 0$, and $w \neq 0$ for $z \neq -\sqrt{2}$.

(a) Insert the four expressions:

$$\begin{aligned} \frac{(1 + \varepsilon^2 w^2)^5 (1 - w^2)^3}{w^4 (w^2 + \varepsilon^2)^4} &= \frac{(4\sqrt{2}\varepsilon^2)^5 (2\sqrt{2}\varepsilon)^3}{\varepsilon^4 (4\sqrt{2}\varepsilon^2)^4} \cdot \frac{N(z) (\sqrt{2} - z)^{-10-6+4+8}}{(\sqrt{2} + z)^4 z^4} \\ &= 128\varepsilon \cdot \frac{N(z)}{z^4 (\sqrt{2} - z)^4 (\sqrt{2} + z)^4} = 128\varepsilon f(z), \end{aligned}$$

because $(4\sqrt{2}\varepsilon^2)(2\sqrt{2}\varepsilon)^3\varepsilon^{-4} = 4\sqrt{2} \cdot 16\sqrt{2} \cdot \varepsilon = 128\varepsilon$ and $(\sqrt{2} - z)(\sqrt{2} + z) = 2 - z^2$.

$$\begin{aligned} \text{(b)} \quad \frac{dw}{dz} &= i\varepsilon \frac{(\sqrt{2} - z) + (\sqrt{2} + z)}{(\sqrt{2} - z)^2} = \frac{2\sqrt{2}i\varepsilon}{(\sqrt{2} - z)^2}, \text{ hence } \frac{2i\varepsilon}{w^2 + \varepsilon^2} \cdot \frac{dw}{dz} = 2i\varepsilon \cdot \frac{2\sqrt{2}i\varepsilon}{(\sqrt{2} - z)^2} \cdot \\ &\frac{(\sqrt{2} - z)^2}{-4\sqrt{2}\varepsilon^2 z} = \frac{1}{z}. \end{aligned}$$

(c) $w(1 + i) = i\varepsilon \frac{\sqrt{2} + 1 + i}{\sqrt{2} - 1 - i} = \frac{i\varepsilon(\sqrt{2} + 1) + i^2\varepsilon}{\varepsilon - i} = \frac{i - \varepsilon}{\varepsilon - i} = -1$, using $\varepsilon(\sqrt{2} + 1) = 1$; in the same way $w(1 - i) = \frac{i + \varepsilon}{\varepsilon + i} = 1$.

Formula (5.9) follows from (a) and (b): $h_n = f^n \cdot \frac{1}{z} = (128\varepsilon)^{-n} \left[\frac{(1 + \varepsilon^2 w^2)^5 (1 - w^2)^3}{w^4 (w^2 + \varepsilon^2)^4} \right]^n \cdot \frac{2i\varepsilon}{w^2 + \varepsilon^2} \frac{dw}{dz}$.

The identity (5.11) and the expansion (5.12). $\varkappa(y + 2\varepsilon x) = (w^2 + \varepsilon^2) + 2\varepsilon(1 - w^2) = (1 - 2\varepsilon)w^2 + (\varepsilon^2 + 2\varepsilon) = \varepsilon^2 w^2 + 1$ by (A.2). Since $1 - w^2 = \varkappa x$ and $w^2 + \varepsilon^2 = \varkappa y$,

$$\begin{aligned} k_n &= \frac{(\varkappa(y + 2\varepsilon x))^{5n} (\varkappa x)^{3n}}{w^{4n} (\varkappa y)^{4n+1}} = \varkappa^{4n-1} \frac{(y + 2\varepsilon x)^{5n} x^{3n}}{y^{4n+1} w^{4n}} \\ &= \varkappa^{4n-1} \sum_{i=0}^{5n} \binom{5n}{i} (2\varepsilon)^i \frac{x^{3n+i} y^{5n-i-(4n+1)}}{w^{4n}}. \end{aligned}$$

Proof of Lemma 5.17. (a) is clear from the definition, since R_2 is a ring containing $\sqrt{2}$. (b) $2^k = (\sqrt{2})^{2k}$; an odd integer m and its inverse are 2-integral (Lemma 5.2); ε and $\varepsilon^{-1} = \sqrt{2} + 1$ lie in $\mathbb{Z}[\sqrt{2}] \subseteq R_2$; and $\varkappa = (\sqrt{2})^3 \varepsilon$.

(c) *Legendre's formula.* The exponent of 2 in $a!$ is $\sum_{k \geq 1} \lfloor a/2^k \rfloor$ (count the multiples of 2, of 4, $\dots$ among $1, \dots, a$). If $a = \sum_i d_i 2^i$ in binary, then $\lfloor a/2^k \rfloor = \sum_{i \geq k} d_i 2^{i-k}$, and summing over $k \geq 1$ gives $\sum_i d_i (2^i - 1) = a - s_2(a)$. *Kummer's formula* for the central binomial coefficient follows: the exponent of 2 in $\binom{2j}{j} = (2j)!/(j!)^2$ is $(2j - s_2(2j)) - 2(j - s_2(j)) = s_2(j)$, because $2j$ and j have the same binary digits. *The inequality* $2s_2(a) \leq a + 1$: it holds for $a = 0, 1$; and $s_2(2q) = s_2(q)$, $s_2(2q + 1) = s_2(q) + 1$ give $2s_2(2q) \leq q + 1 \leq 2q + 1$ and, for $q \geq 1$, $2s_2(2q + 1) \leq q + 3 \leq 2q + 2$. Finally a number $a \geq 1$ has $\lfloor \log_2 a \rfloor + 1$ binary digits, so $s_2(a) \leq \lfloor \log_2 a \rfloor + 1$.

(d) If $x = (\sqrt{2})^{-2L} y = 2^{-L} y$ with $y \in R_2$, then $2^L x = y$ is rational and 2-integral.

Atoms of the first kind: Lemma 5.18(a). For an integer $e \neq -1$ the function w^e has the primitive $w^{e+1}/(e+1)$ on $\mathbb{C} \setminus \{0\}$. For even $e = 2q$ this gives $F(1) - F(-1) = (1 - (-1)^{2q+1})/(2q+1) = 2/(2q+1)$: the function w^{2q} has data $(\frac{2}{2q+1}, 0)$, for negative q as well. Now

$$\frac{x^a}{w^{2m}} = \frac{1}{\varkappa^a} \sum_{i=0}^a (-1)^i \binom{a}{i} w^{2i-2m} \quad \text{has data} \quad \left(\frac{2}{\varkappa^a} \sum_{i=0}^a \frac{(-1)^i \binom{a}{i}}{2i+1-2m}, 0 \right).$$

The sum is evaluated by the following identity with $t = 1 - 2m$.

Lemma A.1. *For every integer $a \geq 0$ and every number t that is not of the form $0, -2, -4, \dots, -2a$,*

$$S_a(t) := \sum_{i=0}^a \frac{(-1)^i \binom{a}{i}}{2i+t} = \frac{2^a a!}{\prod_{j=0}^a (2j+t)}.$$

Proof. Induction on a ; the case $a = 0$ is $1/t = 1/t$. By Pascal's rule $\binom{a+1}{i} = \binom{a}{i} + \binom{a}{i-1}$,

$$S_{a+1}(t) = \sum_i \frac{(-1)^i \binom{a}{i}}{2i+t} + \sum_i \frac{(-1)^i \binom{a}{i-1}}{2i+t} = S_a(t) - S_a(t+2)$$

(put $i = i' + 1$ in the second sum). By the induction hypothesis

$$\begin{aligned} S_a(t) - S_a(t+2) &= 2^a a! \left(\frac{1}{\prod_{j=0}^a (2j+t)} - \frac{1}{\prod_{j=1}^{a+1} (2j+t)} \right) \\ &= 2^a a! \cdot \frac{(2a+2+t) - t}{\prod_{j=0}^{a+1} (2j+t)} = \frac{2^{a+1} (a+1)!}{\prod_{j=0}^{a+1} (2j+t)}. \end{aligned} \quad \square$$

Hence $B_{m,a} = 2^{a+1} a! \kappa^{-a} / \prod_{j=0}^a (2j+1-2m)$. The product in the denominator consists of odd integers, so its inverse has level ≥ 0 ; 2^{a+1} has level $2a+2$; $a!$ has level $2a-2s_2(a)$; and κ^{-a} has level $-3a$. Total: $a+2-2s_2(a)$. For example, for $m = 2$ and $a = 3$: $B_{2,3} = 2^4 \cdot 6 / (\kappa^3 \cdot (-3)(-1)(1)(3)) = \frac{32}{3} \kappa^{-3}$, which is the entry $\frac{32}{3}$ of Example 5.19 after multiplication by κ^3 .

Atoms of the second kind: Lemma 5.18(b). The case $m = 0$. We claim that $(w^2 + \varepsilon^2)^{-(j+1)}$ has data (α_j, β_j) , where $\alpha_0 = 0$, $\beta_0 = 1$ and

$$(A.3) \quad \alpha_{j+1} = \frac{(2j+1)\alpha_j + 2\kappa^{-(j+1)}}{2(j+1)\varepsilon^2}, \quad \beta_{j+1} = \frac{(2j+1)\beta_j}{2(j+1)\varepsilon^2}.$$

For $j = 0$ take $F = 0$ in Definition 5.14. For the step, differentiate and write $w^2 = (w^2 + \varepsilon^2) - \varepsilon^2$:

$$\frac{d}{dw} \frac{w}{(w^2 + \varepsilon^2)^{j+1}} = \frac{1}{(w^2 + \varepsilon^2)^{j+1}} - \frac{2(j+1)w^2}{(w^2 + \varepsilon^2)^{j+2}} = -\frac{2j+1}{(w^2 + \varepsilon^2)^{j+1}} + \frac{2(j+1)\varepsilon^2}{(w^2 + \varepsilon^2)^{j+2}}.$$

Solving for the last term expresses $(w^2 + \varepsilon^2)^{-(j+2)}$ as $\frac{1}{2(j+1)\varepsilon^2}$ times the sum of $(2j+1)(w^2 + \varepsilon^2)^{-(j+1)}$ and an exact derivative. The function $w(w^2 + \varepsilon^2)^{-(j+1)}$ takes the values $\pm(1 + \varepsilon^2)^{-(j+1)} = \pm\kappa^{-(j+1)}$ at $w = \pm 1$, so the exact derivative has data $(2\kappa^{-(j+1)}, 0)$; this gives (A.3).

Closed forms and levels. From $\binom{2j+2}{j+1} = \frac{2(2j+1)}{j+1} \binom{2j}{j}$ one gets by induction

$$\beta_j = \frac{1}{(4\varepsilon^2)^j} \binom{2j}{j}, \quad \alpha_j = \beta_j A_j \quad \text{with} \quad A_0 = 0, \quad A_{j+1} = A_j + \frac{(\sqrt{2})^{j+1} \varepsilon^{j-1}}{2(2j+1) \binom{2j}{j}}.$$

(For the second formula divide (A.3) by β_{j+1} : the new term is $\kappa^{-(j+1)} / ((j+1)\varepsilon^2 \beta_{j+1})$, and $\kappa^{-(j+1)} 4^{j+1} = (\sqrt{2})^{j+1} \varepsilon^{-(j+1)}$, $(j+1) \binom{2j+2}{j+1} = 2(2j+1) \binom{2j}{j}$.) By Kummer's formula β_j has level $2s_2(j) - 4j \geq -4j$. The increment of A has level $(j+1) - 2 - 2s_2(j) \geq -2$ because $2s_2(j) \leq j+1$; so $A_j \in \mathfrak{L}_{-2}$ and, for $j \geq 1$, α_j has level at least $2s_2(j) - 4j - 2 \geq -4j$. Thus

$$(A.4) \quad \alpha_j, \beta_j \in \mathfrak{L}_{-4j} \quad (j \geq 0).$$

General m . We show by induction on m that for all $k \geq 1$ the function $w^{-2m}(w^2 + \varepsilon^2)^{-k}$ has data in $\mathfrak{L}_{4-4k}$. For $m = 0$ this is (A.4) with $j = k - 1$. For the step we use the partial fraction identity

$$\frac{1}{w^{2(m+1)}(w^2 + \varepsilon^2)^k} = \frac{1}{\varepsilon^2} \left(\frac{1}{w^{2(m+1)}(w^2 + \varepsilon^2)^{k-1}} - \frac{1}{w^{2m}(w^2 + \varepsilon^2)^k} \right)$$

(put the right-hand side over a common denominator: the numerator is $(w^2 + \varepsilon^2) - w^2 = \varepsilon^2$) and a second induction on k . The second function on the right has data in $\mathfrak{L}_{4-4k}$ by the induction hypothesis for m . The first one has data in $\mathfrak{L}_{4-4(k-1)} \subseteq \mathfrak{L}_{4-4k}$ if $k \geq 2$; if $k = 1$ it is $w^{-2(m+1)}$, with data $(\frac{2}{-2m-1}, 0)$ of level $2 \geq 0$. Multiplication by ε^{-2} does not change levels.

Finally $y^{-k}w^{-2m} = \varkappa^k w^{-2m}(w^2 + \varepsilon^2)^{-k}$ and $\varkappa^k \in \mathfrak{L}_{3k}$: the data have level at least $3k + 4 - 4k = 4 - k$. This proves Lemma 5.18(b).

The bookkeeping, once more. After multiplication by $-2\varepsilon(128\varepsilon)^{-n}$ and $(128\varepsilon)^{-n}$ (levels $2 - 14n$ and $-14n$), Table 5 becomes the following table of the levels of the contributions to u_n and to c_n (recall that level k means divisibility by $(\sqrt{2})^k$; for a rational number, level $2k$ means divisibility by 2^k).

terms	contribution to u_n	contribution to c_n
$0 \leq i \leq n - 1$	$\geq n - 1 + 2i - 2L$	none
$n \leq i \leq 5n$, atoms $x^b w^{-4n}$	$\geq 2(i - n)$	none
$n \leq i \leq 5n$, atoms $y^{-k} w^{-4n}$	$\geq i - n + 2$	$\geq i - n$

All entries of the column for u_n are $\geq -2L$ and those of the column for c_n are ≥ 0 , which is Proposition 5.12. The only negative levels come from the first row, that is from the binary digit sums $s_2(a)$ with $3n \leq a \leq 4n - 1$ in Lemma 5.18(a); this is the origin of the factor $2^{\lfloor \log_2(4n) \rfloor}$.

Example 5.19 by hand: the columns $i = 0$ and $i = 1$. For $n = 1$ the terms of (5.12) are $\varkappa^3 \binom{5}{i} (2\varepsilon)^i T_i$ with $T_i = x^{3+i} y^{-i} w^{-4}$, and Lemma 5.18 is used with $m = 2$.

The column $i = 0$. $T_0 = x^3 w^{-4}$ is an atom of the first kind, with data $(B_{2,3}, 0) = (\frac{32}{3} \varkappa^{-3}, 0)$. Multiplication by $\varkappa^3$ gives the contribution $(\frac{32}{3}, 0)$.

The column $i = 1$. Here $T_1 = x^4 y^{-1} w^{-4}$. Since $y = 1 - x$, we have $x^4 - 1 = -(1 + x + x^2 + x^3)(1 - x) = -(1 + x + x^2 + x^3)y$, hence

$$T_1 = \frac{1}{y w^4} - \frac{1 + x + x^2 + x^3}{w^4}.$$

The second part consists of atoms of the first kind, with

$$B_{2,0} = -\frac{2}{3}, \quad B_{2,1} = \frac{4}{3\varkappa}, \quad B_{2,2} = \frac{16}{3\varkappa^2}, \quad B_{2,3} = \frac{32}{3\varkappa^3}.$$

For the first part we use the partial fraction identity twice, starting from the data $(0, 1)$ of $(w^2 + \varepsilon^2)^{-1}$, $(-2, 0)$ of w^{-2} and $(-\frac{2}{3}, 0)$ of w^{-4} :

$$\begin{aligned} \frac{1}{w^2(w^2 + \varepsilon^2)} &= \frac{1}{\varepsilon^2} \left(\frac{1}{w^2} - \frac{1}{w^2 + \varepsilon^2} \right) && \text{has data } \left(-\frac{2}{\varepsilon^2}, -\frac{1}{\varepsilon^2} \right), \\ \frac{1}{w^4(w^2 + \varepsilon^2)} &= \frac{1}{\varepsilon^2} \left(\frac{1}{w^4} - \frac{1}{w^2(w^2 + \varepsilon^2)} \right) && \text{has data } \left(\frac{2}{\varepsilon^4} - \frac{2}{3\varepsilon^2}, \frac{1}{\varepsilon^4} \right). \end{aligned}$$

As $y^{-1} = \varkappa(w^2 + \varepsilon^2)^{-1}$, the function T_1 has data (U_1, V_1) with

$$U_1 = \varkappa \left(\frac{2}{\varepsilon^4} - \frac{2}{3\varepsilon^2} \right) + \frac{2}{3} - \frac{4}{3\varkappa} - \frac{16}{3\varkappa^2} - \frac{32}{3\varkappa^3}, \quad V_1 = \frac{\varkappa}{\varepsilon^4}.$$

Now multiply by $\varkappa^3 \binom{5}{1}(2\varepsilon) = 10\varepsilon \varkappa^3$ and use $\varkappa = 2\sqrt{2}\varepsilon$, so that $\varkappa^2 = 8\varepsilon^2$, $\varkappa^3 = 16\sqrt{2}\varepsilon^3$ and $\varkappa^4 = 64\varepsilon^4$:

$$10\varepsilon \varkappa^3 V_1 = \frac{10\varkappa^4}{\varepsilon^3} = 640\varepsilon,$$

$$10\varepsilon \varkappa^3 U_1 = 10\varepsilon \left(128 - \frac{128}{3}\varepsilon^2 + \frac{32\sqrt{2}}{3}\varepsilon^3 - \frac{32}{3}\varepsilon^2 - \frac{32\sqrt{2}}{3}\varepsilon - \frac{32}{3} \right) = 10\varepsilon \cdot \frac{128}{3} (1 + \sqrt{2}) = \frac{1280}{3}.$$

For the last line: $\varepsilon^3 - \varepsilon = -2\varepsilon^2$ (because $\varepsilon^2 = 1 - 2\varepsilon$), so the bracket equals $\frac{352}{3} - \frac{\varepsilon^2}{3}(160 + 64\sqrt{2})$; and $\varepsilon^2(160 + 64\sqrt{2}) = (3 - 2\sqrt{2})(160 + 64\sqrt{2}) = 224 - 128\sqrt{2}$, which gives $\frac{1280}{3}(1 + \sqrt{2})$; finally $\varepsilon(1 + \sqrt{2}) = 1$. These are the entries $\frac{1280}{3}$ and 640 of the table. The remaining four columns are obtained in the same way, with more terms.

Why every entry is rational. In the variable z the identity (5.11) is the relation $W - 3 = (W - 2) - 1$. Indeed, by the formulas in the proof of Lemma 5.13,

$$\frac{2\varepsilon x}{W - 2} = -y = \frac{y + 2\varepsilon x}{W - 3} = \frac{2\varepsilon z}{(\sqrt{2} - z)^2}.$$

So, up to the common factor $(2\varepsilon z/(\sqrt{2} - z)^2)^{5n}$, the binomial expansion of $(y + 2\varepsilon x)^{5n}$ in (5.12) is the expansion $(W - 3)^{5n} = \sum_i \binom{5n}{i} (-1)^{5n-i} (W - 2)^i$, and the i -th term of (5.12) corresponds, under the transfer of Lemma 5.15, to the function

$$(-1)^{n-i} \binom{5n}{i} \frac{(W - 2)^{3n+i}}{Y^{4n}} \cdot \frac{1}{z},$$

whose sum over i is h_n . This function has rational coefficients, so Proposition 4.6 with $R = \mathbb{Q}$ gives it a constant c and a primitive with rational data, and the argument of Lemma 4.2 (which applies to any function in place of h_n) shows that the w -computation must give the same numbers. Write $(U'_i, V'_i) := \varkappa^{4n-1} \binom{5n}{i} (2\varepsilon)^i (U_i, V_i)$ for the contribution of the i -th term to the data of k_n , where (U_i, V_i) are the data of T_i as in the proof of Proposition 5.12; for $n = 1$ the numbers U'_i and V'_i/ε are the entries of the table in Example 5.19. Then the i -th term contributes the rational numbers $-2\varepsilon(128\varepsilon)^{-n} U'_i$ to u_n and $(128\varepsilon)^{-n} V'_i$ to c_n . For $n = 1$ these are $-U'_i/64$ and $V'_i/(128\varepsilon)$; for instance formula (4.6), applied to $P = 5(W - 2)^4$ and $m = 2$, gives $c = 5 \cdot \frac{4 \cdot 2}{192} \binom{4}{2} 2^2 = 5 = 640/128$ for $i = 1$, and the six contributions to c_1 are $0 + 5 + 100 + 800 + 2800 + 3640 = 7345$.

A.6. Growth.

Fekete's lemma. Let $(a_n)_{n \geq 1}$ be real numbers with $a_{m+n} \geq a_m + a_n$ for all m, n , and suppose $a_n \leq Bn$ for all n . Then a_n/n converges to $\sup_n a_n/n$.

Proof. Let $r = \sup_n a_n/n \leq B$. Fix m and write $n = qm + t$ with $0 \leq t < m$. By superadditivity $a_n \geq qa_m + a_t$ (with $a_0 := 0$), so $\frac{a_n}{n} \geq \frac{qm}{n} \cdot \frac{a_m}{m} + \frac{a_t}{n}$. As $n \rightarrow \infty$, $qm/n \rightarrow 1$ and $a_t/n \rightarrow 0$ (t takes finitely many values); hence $\liminf a_n/n \geq a_m/m$ for every m , that is $\liminf a_n/n \geq r \geq \limsup a_n/n$. $\square$

We apply it to $a_n = \log c_n$: superadditivity is $c_{m+n} \geq c_m c_n$, and $a_n \leq n \log f(-x)$.

Supermultiplicativity. If $F = \sum_k a_k z^k$ and $G = \sum_k b_k z^k$ have non-negative coefficients, the coefficient of z^{a+b} in FG is $\sum_{i+j=a+b} a_i b_j \geq a_a b_b$. With $F = \tilde{P}^m$, $G = \tilde{P}^n$, $a = 4m$, $b = 4n$ this is $c_{m+n} \geq c_m c_n$. The coefficients of $\tilde{P}$ are non-negative because $\tilde{N}$ is a product of polynomials with positive coefficients and $(2 - z^2)^{-1} = \frac{1}{2} \sum_{k \geq 0} (z^2/2)^k$.

The upper bound. For $0 < x < \sqrt{2}$ the power series $\tilde{P}(x)^n = \sum_k \gamma_k x^k$ (where $\gamma_k \geq 0$ is the coefficient of z^k in $\tilde{P}^n$) converges, and all its terms are non-negative; hence $c_n x^{4n} = \gamma_{4n} x^{4n} \leq \tilde{P}(x)^n$, that is $c_n \leq (\tilde{P}(x)/x^4)^n = f(-x)^n$.

The numbers. For $x = \frac{3913}{10000}$,

$$f(-x) = \frac{((x+1)(x+2))^5 (x^2 + 2x + 2)^3}{x^4 (2 - x^2)^4} = 37810.99167 \dots < 37811$$

(exact rational arithmetic). And $e^{10.540357} = (e)^{10} \cdot e^{0.540357} \geq (2.7182818283)^{10} \cdot \sum_{i=0}^9 \frac{0.540357^i}{i!} = 22026.4657 \dots \times 1.7166195 \dots = 37811.0625 \dots > 37811$.

A.7. The polygon estimate.

$|f|^2$ as a quotient of polynomials. For $z = x + iy$: $|z-1|^2 = (x-1)^2 + y^2$, $|z-2|^2 = (x-2)^2 + y^2$, $|z^2 - 2z + 2|^2 = |z-1-i|^2 |z-1+i|^2 = ((x-1)^2 + (y-1)^2)((x-1)^2 + (y+1)^2)$, $|z|^2 = x^2 + y^2$ and $|z^2 - 2|^2 = (x^2 - y^2 - 2)^2 + 4x^2 y^2$. Hence (7.1) holds with

$$\begin{aligned} \text{Num} &= \left[((x-1)^2 + y^2)((x-2)^2 + y^2) \right]^5 \left[((x-1)^2 + (y-1)^2)((x-1)^2 + (y+1)^2) \right]^3, \\ \text{Den} &= (x^2 + y^2)^4 ((x^2 - y^2 - 2)^2 + 4x^2 y^2)^4. \end{aligned}$$

The two upper segments of Γ are

$$\begin{aligned} 1 \rightarrow z_s : \quad x &= 1 + \frac{3403}{16384} t, \quad y = \frac{32371}{65536} t, \\ z_s \rightarrow 1+i : \quad x &= \frac{19787}{16384} - \frac{3403}{16384} t, \quad y = \frac{32371}{65536} + \frac{33165}{65536} t \quad (0 \leq t \leq 1). \end{aligned}$$

Each of the four quadratic factors of Num becomes a polynomial of degree 2 in t , so Num has degree $2(5 + 5 + 3 + 3) = 32$; in Den the factor $x^2 + y^2$ has degree 2 and the factor $(x^2 - y^2 - 2)^2 + 4x^2 y^2$ has degree 4 in t , so Den has degree $4 \cdot 2 + 4 \cdot 4 = 24$; and $Q = \rho^2 \text{Den} - \text{Num}$ has degree 32.

Bernstein certificates. Every polynomial Q of degree $\leq d$ can be written in exactly one way as $Q(s) = \sum_{j=0}^d \beta_j s^j (1-s)^{d-j}$ (the polynomials $\binom{d}{j} s^j (1-s)^{d-j}$ are the *Bernstein polynomials* of degree d ; we absorb the binomial coefficient into β_j). If $Q(s) = \sum_i a_i s^i$, then, multiplying s^i by $1 = (s + (1-s))^{d-i}$,

$$\beta_j = \sum_{i=0}^j \binom{d-i}{j-i} a_i.$$

If all β_j are ≥ 0 , then $Q(s) \geq 0$ for $0 \leq s \leq 1$, since $s^j (1-s)^{d-j} \geq 0$ there. The converse is false in general, but for a polynomial that is positive on an interval the coefficients become positive after the interval is cut into sufficiently small pieces. Note that $\beta_0 = Q(0)$ and $\beta_d = Q(1)$. A list of non-negative β_j together with the polynomial identity is a certificate of non-negativity that can be checked with exact rational arithmetic and nothing else.

In our case $d = 32$. For the first segment the certificate for $[0, 1]$ has $\beta_0 = Q(0) = \rho^2 = 2.128 \dots \cdot 10^{-6}$ (at $z = 1$, where $f = 0$ and $\text{Den} = 1$), and its smallest coefficient is $\beta_{32} = Q(1) = 7.5 \dots \cdot 10^{-10}$ (at z_s). For the second segment the expansion on $[0, 1]$ has one negative coefficient ($\beta_1 = -1.0 \dots \cdot 10^{-8}$), so the interval is halved: with $Q(\frac{1}{2}s)$ and $Q(\frac{1}{2} + \frac{1}{2}s)$ all coefficients are positive, the smallest being $7.5 \dots \cdot 10^{-10}$ and $5.4 \dots \cdot 10^{-3}$. The small margin at z_s reflects the fact that ρ^2 exceeds $|f(z_s)|^2$ only by $5.2 \cdot 10^{-12}$. The denominator is positive on Γ because Γ avoids 0 and $\pm\sqrt{2}$.

The inequality $\rho < e^{-6.530134}$. Equivalently $e^{6.530134} < 1/\rho = 685.49069\dots$. We have $e < 2.7182818286$, and for $0 \leq x \leq 1$ the exponential series satisfies $e^x \leq \sum_{m=0}^{11} \frac{x^m}{m!} + \frac{13x^{12}}{12 \cdot 12!}$. Hence

$$\begin{aligned} e^{6.530134} &= e^6 \cdot e^{0.530134} < (2.7182818286)^6 \times 1.69915999 \\ &= 403.42879\dots \times 1.69915999 < 685.49007 < \frac{1}{\rho}. \end{aligned}$$

The lengths. For a complex number $a + bi$, $|a + bi| \leq |a| + |b|$. So $|z_s - 1| \leq \frac{3403}{16384} + \frac{32371}{65536} < 0.71$ and $|1 + i - z_s| \leq \frac{3403}{16384} + \frac{33165}{65536} < 0.72$: the four segments of Γ are shorter than 1. (Their true lengths are $0.5358\dots$ and $0.5470\dots$, so the constant 4 in Proposition 7.3 could be replaced by 2.17; this is irrelevant.)

A.8. From the prime number theorem to the multiplier.

Chebyshev's functions. Let $N \geq 1$ be an integer. The exponent of a prime p in $\text{lcm}(1, \dots, N)$ is the largest k with $p^k \leq N$, which is also the number of $k \geq 1$ with $p^k \leq N$. Hence $\log \text{lcm}(1, \dots, N) = \sum_{p^k \leq N} \log p = \psi(N)$. Moreover

$$0 \leq \psi(x) - \theta(x) = \sum_{p \leq \sqrt{x}} ([\log_p x] - 1) \log p \leq \sum_{p \leq \sqrt{x}} \log x \leq \sqrt{x} \log x,$$

because only primes with $p^2 \leq x$ contribute and $[\log_p x] \log p \leq \log x$. So $\theta(x) \sim x$ implies $\psi(4n) = 4n + o(n)$.

Proof of Lemma 8.1. Write $n = pk + r_0$ with $0 \leq r_0 < p$ and put $t = r_0/p = \{n/p\}$. For $m \in \{3, 4, 5\}$ we have $mn = p \cdot mk + p \cdot mt$, so

$$\left\lfloor \frac{mn}{p} \right\rfloor = mk + [mt], \quad \frac{mn \bmod p}{p} = \{mt\}.$$

Thus p (with $p^2 > 4n$) is removable if and only if $[4t]$ is even and $\{5t\} + \{3t\} < \{4t\}$. Now $\{5t\} + \{3t\} - \{4t\} = 4t - c(t)$ with $c(t) = [5t] + [3t] - [4t]$, so the inequality reads $4t < c(t)$. The integer parts change only at the multiples of $\frac{1}{5}, \frac{1}{3}, \frac{1}{4}$:

$t \in$	$[0, \frac{1}{5})$	$[\frac{1}{5}, \frac{1}{4})$	$[\frac{1}{4}, \frac{1}{3})$	$[\frac{1}{3}, \frac{2}{5})$	$[\frac{2}{5}, \frac{1}{2})$	$[\frac{1}{2}, \frac{3}{5})$	$[\frac{3}{5}, \frac{2}{3})$	$[\frac{2}{3}, \frac{3}{4})$	$[\frac{3}{4}, \frac{4}{5})$	$[\frac{4}{5}, 1)$
$[5t], [3t], [4t]$	0, 0, 0	1, 0, 0	1, 0, 1	1, 1, 1	2, 1, 1	2, 1, 2	3, 1, 2	3, 2, 2	3, 2, 3	4, 2, 3
$c(t)$	0	1	0	1	2	1	2	3	2	3
$4t < c(t)?$	no	yes	no	no	yes	no	no	yes	no	no
$[4t]$ even?	yes	yes	no	no	no	yes	yes	yes	no	no

(For instance on $[\frac{1}{3}, \frac{2}{5})$ we have $4t \geq \frac{4}{3} > 1 = c$, and on $[\frac{2}{5}, \frac{1}{2})$ we have $4t < 2 = c$ but $[4t] = 1$ is odd.) Both conditions hold exactly on $[\frac{1}{5}, \frac{1}{4}) \cup [\frac{2}{3}, \frac{3}{4})$.

The windows. $k + \frac{2}{3} \leq \frac{n}{p} < k + \frac{3}{4}$ is equivalent to $\frac{4n}{4k+3} < p \leq \frac{3n}{3k+2}$, and $k + \frac{1}{5} \leq \frac{n}{p} < k + \frac{1}{4}$ to $\frac{4n}{4k+1} < p \leq \frac{5n}{5k+1}$; for $k = 0$ the latter window consists of numbers $p > 4n$, which do not divide $\text{lcm}(1, \dots, 4n)$. Different windows are disjoint, since they correspond to disjoint sets of values of n/p .

Proof of Proposition 8.3. (a) For fixed k , the prime number theorem gives

$$\begin{aligned} \sum_{\frac{4n}{4k+3} < p \leq \frac{3n}{3k+2}} \log p &= \theta\left(\frac{3n}{3k+2}\right) - \theta\left(\frac{4n}{4k+3}\right) \\ &= \left(\frac{3}{3k+2} - \frac{4}{4k+3}\right)n + o(n) = \frac{n}{(3k+2)(4k+3)} + o(n), \end{aligned}$$

and in the same way the window of the second kind contributes $\frac{n}{(4k+1)(5k+1)} + o(n)$. Summing over the $2K-1$ windows with $k < K$ gives $\log \Phi_n^{(K)} = \varpi_K n + o(n)$, and $\log D_n^{(K)} = \psi(4n) - \log \Phi_n^{(K)} = (4 - \varpi_K)n + o(n)$. A prime p in a window with $k < K$ satisfies $p > \frac{4n}{4K} = \frac{n}{K}$. If $n \geq 4K^2$, then $\frac{n}{K} \geq 2\sqrt{n}$, so $p^2 > 4n$ and p is removable by Lemma 8.1; also $p \leq \frac{3n}{2} \leq 4n$. Hence $\Phi_n^{(K)}$ divides Φ_n , and D_n divides $D_n^{(K)}$.

(b) Let $n \geq 4K^2$. A prime that divides Φ_n but not $\Phi_n^{(K)}$ lies in a window with $k \geq K$, so it is at most $\max\left(\frac{3n}{3K+2}, \frac{5n}{5K+1}\right) < \frac{n}{K}$. Therefore

$$\log \Phi_n^{(K)} \leq \log \Phi_n \leq \log \Phi_n^{(K)} + \theta\left(\frac{n}{K}\right),$$

and, dividing by n and using (a) and $\theta(n/K) = n/K + o(n)$,

$$\varpi_K \leq \liminf_{n \rightarrow \infty} \frac{\log \Phi_n}{n} \leq \limsup_{n \rightarrow \infty} \frac{\log \Phi_n}{n} \leq \varpi_K + \frac{1}{K}.$$

Letting $K \rightarrow \infty$ gives $\frac{1}{n} \log \Phi_n \rightarrow \varpi$ and $\frac{1}{n} \log D_n \rightarrow 4 - \varpi$.

(c) *The value of ϖ .* With the digamma function $\Psi = \Gamma'/\Gamma$ (the logarithmic derivative of Euler's Gamma function; this Γ is not the polygon of Section 7) one has $\sum_{k \geq 0} \left(\frac{1}{k+a} - \frac{1}{k+b}\right) = \Psi(b) - \Psi(a)$ for $a, b > 0$. Since $\frac{1}{(3k+2)(4k+3)} = \frac{1}{k+2/3} - \frac{1}{k+3/4}$ and $\frac{1}{(4k+1)(5k+1)} = \frac{1}{k+1/5} - \frac{1}{k+1/4}$,

$$\varpi = \Psi\left(\frac{3}{4}\right) - \Psi\left(\frac{2}{3}\right) + \Psi\left(\frac{5}{4}\right) - \Psi\left(\frac{6}{5}\right) = 0.2323735360\dots + 0.0615863632\dots = 0.2939598992\dots$$

The first terms are $\frac{1}{6} + \frac{1}{35} + \frac{1}{30} + \frac{1}{88} + \frac{1}{99} + \dots$.

The lower bound for $K = 10^5$. A direct summation gives $\varpi_{10^5} = 0.29395856588\dots$ (the difference $\varpi - \varpi_K$ is about $\frac{2}{15K}$). A proof that needs only a small computation, which is the one formalised, goes as follows. The first 20 terms of the first sum and the first 19 terms of the second sum add up to $0.28730270\dots$ (exact rational arithmetic). For $x \geq 20$,

$$\begin{aligned} \frac{1}{(3x+2)(4x+3)} &\geq \frac{1}{12} \left(\frac{1}{x + \frac{43}{200}} - \frac{1}{x+1 + \frac{43}{200}} \right), \\ \frac{1}{(4x+1)(5x+1)} &\geq \frac{1}{20} \left(\frac{1}{x - \frac{67}{250}} - \frac{1}{x+1 - \frac{67}{250}} \right) \end{aligned}$$

(clear denominators: the first inequality is $12(x+a)(x+a+1) \geq (3x+2)(4x+3)$ with $a = \frac{43}{200}$, i.e. $0.16x \geq 6 - 12a(a+1) = 2.86\dots$, and the second is $0.28x \geq 1 + 20b(1-b) = 4.92\dots$ with $b = \frac{67}{250}$). The right-hand sides telescope when summed over $x = 20, \dots, K-1$:

$$\varpi_K \geq 0.28730270 + \frac{1}{12} \left(\frac{1}{20.215} - \frac{1}{K+0.215} \right) + \frac{1}{20} \left(\frac{1}{19.732} - \frac{1}{K-0.268} \right),$$

which for $K = 10^5$ equals $0.2939576\dots \geq 0.293957$.

A.9. Hata's lemma. We prove Lemma 1.2 in the following form, which is the one that has been formalised: *let θ be irrational, let $\sigma, \tau > 0$, and suppose that for every $\eta > 0$ there is n_1 such that for all $n \geq n_1$*

$$(A.5) \quad e^{(\sigma-\eta)n} \leq |q_n| \leq e^{(\sigma+\eta)n} \quad \text{and} \quad |q_n\theta - p_n| \leq e^{-(\tau-\eta)n}.$$

Then for every $\mu > 1 + \sigma/\tau$ there is b_0 with $|\theta - a/b| > b^{-\mu}$ for all $a \in \mathbb{Z}$ and all integers $b \geq b_0$. The hypotheses of Lemma 1.2 imply (A.5) by the definition of limit and upper limit.

Proof. Fix η with $0 < 4\eta < \tau$ and the corresponding n_1 , and put $\delta_n := q_n\theta - p_n$. Let a/b be a fraction with $b \geq 1$ and $\Delta := |\theta - a/b|$; then $\Delta > 0$ since θ is irrational. If $\Delta \geq 1$, then $\Delta > b^{-\mu}$ for every $\mu > 0$ and every $b \geq 2$, and there is nothing to prove; so let $\Delta < 1$. Define

$$\ell_1 := \frac{\log(2b)}{\tau - \eta}, \quad \ell_2 := \frac{\log(1/\Delta)}{\sigma + \tau - 2\eta} + 1, \quad N := n_1 + \lceil \max(\ell_1, \ell_2) \rceil.$$

Then $N \geq n_1$ and $\max(\ell_1, \ell_2) \leq N \leq n_1 + \max(\ell_1, \ell_2) + 1$.

(A) Since $N \geq \ell_1$: $e^{-(\tau-\eta)N} \leq \frac{1}{2b}$, hence $b|\delta_N| \leq \frac{1}{2}$.

(B) Since $N \geq \ell_2$: $(\sigma + \tau - 2\eta)N > \log(1/\Delta)$, hence $e^{-(\sigma+\tau-2\eta)N} < \Delta$.

(C) The integer $q_N a - p_N b$ is not zero. Indeed, if $q_N a = p_N b$, then $b\delta_N = bq_N\theta - q_N a = q_N(b\theta - a)$, so $|\delta_N| = |q_N|\Delta$ and

$$\Delta = \frac{|\delta_N|}{|q_N|} \leq \frac{e^{-(\tau-\eta)N}}{e^{(\sigma-\eta)N}} = e^{-(\sigma+\tau-2\eta)N} < \Delta$$

by (B), a contradiction. (This is the only place where the *lower* bound for $|q_n|$ is used.)

(D) From $q_N a - p_N b = b\delta_N - q_N(b\theta - a)$ and (A), (C): $1 \leq |q_N a - p_N b| \leq b|\delta_N| + |q_N|b\Delta \leq \frac{1}{2} + |q_N|b\Delta$. Therefore

$$\Delta \geq \frac{1}{2b|q_N|} \geq \frac{e^{-(\sigma+\eta)N}}{2b} \geq \frac{e^{-(\sigma+\eta)(n_1+1)}}{2b} \cdot e^{-(\sigma+\eta)\max(\ell_1, \ell_2)}.$$

(The last step uses $N \leq n_1 + 1 + \max(\ell_1, \ell_2)$ and $\sigma + \eta > 0$.)

Case $\ell_1 \geq \ell_2$. Then $e^{-(\sigma+\eta)\ell_1} = (2b)^{-(\sigma+\eta)/(\tau-\eta)}$ and $\Delta \geq c' b^{-\kappa_1}$ with $\kappa_1 = 1 + \frac{\sigma+\eta}{\tau-\eta}$ and a constant $c' > 0$ depending only on σ, τ, η, n_1 .

Case $\ell_2 > \ell_1$. Then $e^{-(\sigma+\eta)\ell_2} = e^{-(\sigma+\eta)\Delta^\beta}$ with $\beta = \frac{\sigma+\eta}{\sigma+\tau-2\eta} < 1$, so $\Delta^{1-\beta} \geq c b^{-1}$ and $\Delta \geq c'' b^{-\kappa_2}$ with $\kappa_2 = \frac{1}{1-\beta} = \frac{\sigma+\tau-2\eta}{\tau-3\eta}$.

In both cases $\Delta \geq c b^{-\kappa(\eta)}$ with $c > 0$ independent of a/b and $\kappa(\eta) = \max(\kappa_1, \kappa_2) \leq \frac{\sigma+\tau}{\tau-3\eta}$. As $\eta \rightarrow 0$, $\kappa(\eta) \rightarrow 1 + \frac{\sigma}{\tau}$. Given $\mu > 1 + \sigma/\tau$, choose η so small that $\kappa(\eta) < \mu$; then $c b^{-\kappa(\eta)} > b^{-\mu}$ as soon as $b^{\mu-\kappa(\eta)} > 1/c$, that is for all $b \geq b_0$, with a suitable $b_0 \geq 2$. $\square$

The final arithmetic. With $\sigma \leq 14.2464$ and $\tau \geq 2.824091$ we have $\sigma/\tau \leq 14.2464/2.824091$. Now $2.824091 \times 5.0446 = 14.2464094586 > 14.2464$, so $\sigma/\tau < 5.0446$ and $1 + \sigma/\tau < 6.0446$. (With the weaker bound $\tau \geq 2.82409$ quoted in the introduction: $2.82409 \times 5.0446 = 14.246404414 > 14.2464$.) Since Definition 1.1 takes an infimum, the statement “ $|\pi - a/b| > b^{-\mu}$ for all large b , for every $\mu > 6.0446$ ” is exactly $\mu(\pi) \leq 6.0446$.

APPENDIX B. OTHER MEMBERS OF THE FAMILY

This appendix and the next two are addressed to specialists. Their content is taken from the original draft of this work, where it is treated by the classical route (residues, Laurent expansions, valuations in $\mathbb{Q}(\sqrt{2})$). *It is not part of the formal development*, and in this paper only the triple $(5, 3, 4)$ is proved. The numbers have been recomputed.

B.1. The family. Let e_1, e_2, d be positive integers with $a := e_1 + e_2 - d \geq 1$, and put

$$f_{e_1, e_2, d}(z) = \frac{((z-1)(z-2))^{e_1} (z^2 - 2z + 2)^{e_2}}{z^a (z^2 - 2)^d} = \frac{(W-3)^{e_1} (W-2)^{e_2}}{Y^d}, \quad M := \max(a, d).$$

The computation of Lemma 3.1 gives $f_{e_1, e_2, d}(2/z) = (-1)^d f_{e_1, e_2, d}(z)$. For every $n \geq 1$ with dn even let $J_n = -i \int_{1-i}^{1+i} f_{e_1, e_2, d}(z)^n dz/z$. The integral (1.2) is the case $(e_1, e_2, d) = (5, 3, 4)$.

Claim B.1 (statement of the original draft; not proved in this paper and not formalised). *Let dn be even.*

(i) $J_n = u_n + v_n \pi$ with $u_n \in \mathbb{Q}$ and $v_n = \frac{1}{2} c_n$, where $c_n \in \mathbb{Z}[\frac{1}{2}]$ is the coefficient of z^{an} in the power series of $((z-1)(z-2))^{e_1 n} (z^2 - 2z + 2)^{e_2 n} (z^2 - 2)^{-dn}$.

(ii) Let Φ_n be the product of all primes p with $\sqrt{Mn} < p \leq Mn$ such that $\lfloor dn/p \rfloor$ is even and $\{e_1 n/p\} + \{e_2 n/p\} < \{dn/p\}$. Let $g_n := \max(0, \lceil (d - \frac{3}{2}e_2)n \rceil)$ if $e_2 \leq d$, and $g_n := 1 - \lceil (e_2 - d)n/2 \rceil$ if $e_2 > d$. Then, for $Mn \geq 2$, the number $D_n := 2^{g_n} \text{lcm}(1, \dots, Mn)/\Phi_n$ satisfies $D_n u_n \in \mathbb{Z}$ and $D_n v_n \in \mathbb{Z}$.

(iii) Let $r := \min_{0 < x < \sqrt{2}} \log |f_{e_1, e_2, d}(-x)|$; let $s := -\log \sup_{\Gamma} |f_{e_1, e_2, d}|$ for a piecewise smooth path Γ from $1-i$ to $1+i$ in Ω that is homotopic to the segment in $\mathbb{C} \setminus \{0\}$; let

$$S := \{x > 0 : \lfloor dx \rfloor \text{ even, } \{e_1 x\} + \{e_2 x\} < \{dx\}\},$$

$$\varpi := \int_{S \cap [1/M, \infty)} \frac{dx}{x^2}, \quad C := M + \gamma \log 2 - \varpi,$$

with $\gamma := \max(0, d - \frac{3}{2}e_2)$ if $e_2 \leq d$ and $\gamma := -\frac{1}{2}(e_2 - d)$ if $e_2 > d$. If $s > C$, then $\mu(\pi) \leq 1 + \frac{r+C}{s-C}$.

For $(5, 3, 4)$ one has $a = d = M = 4$, $\gamma = 0$, $g_n = 0$, $S = \bigcup_{k \geq 0} ([k + \frac{1}{5}, k + \frac{1}{4}) \cup [k + \frac{2}{3}, k + \frac{3}{4}))$, and the claim reduces to Theorems 1.4 and 1.5. The arguments of Sections 4, 5.2 and 5.3 use only the shape $P(W)Y^{-2m}/z$ of the integrand and extend to the general case with $5n, 3n, 4n$ replaced by $e_1 n, e_2 n, dn$ and the bound $4n$ for the integration indices replaced by Mn ; the growth is treated as in Section 6. The contribution of the prime 2 depends on the triple: no extra power of 2 is needed when $e_2 \leq d \leq \frac{3}{2}e_2$ (then $g_n = 0$, as for $(5, 3, 4)$), the factor 2^{g_n} is needed when $d > \frac{3}{2}e_2$, and there is a gain when $e_2 > d$; the general statement (ii) is proved in the draft by a valuation count in the coordinate w of Section 5.4. The path Γ must be found, and certified, for each triple.

B.2. Critical points and critical values. The logarithmic derivative of $f_{e_1, e_2, d}$ vanishes where $\frac{e_1}{W-3} + \frac{e_2}{W-2} - \frac{dW}{W^2-8} = 0$, that is where

$$aW^3 - (2e_1 + 3e_2 - 5d)W^2 - (8e_1 + 8e_2 + 6d)W + 16e_1 + 24e_2 = 0.$$

For $(5, 3, 4)$ this is $4W^3 + W^2 - 88W + 152 = 0$ (Remark 7.4). For even d the function f is a rational function of W , and its three critical values are the roots of the resultant of the critical cubic and $(W-3)^{e_1}(W-2)^{e_2} - \lambda(W^2-8)^{d/2}$; for $(5, 3, 4)$ this resultant is, up to a constant factor,

$$2^{20}\lambda^3 - 39647694848\lambda^2 - 115523712\lambda - 3^3 \cdot 5^5,$$

with the roots $\lambda_3 = 37810.99119500591\dots = e^r$ and $-0.00145687795\dots \pm 0.00007500089\dots i$, of modulus e^{-s} and argument $\pm 0.98363\pi$. (The argument explains the signs: heuristically, J_n behaves like $e^{-sn}n^{-1/2} \cos(0.98363\pi n + \varphi_0)$ times a constant, with a constant phase φ_0 , so the sign of J_n alternates, with a break about every $1/(1 - 0.98363) \approx 61$ steps. Indeed, for $n \leq 420$ the numbers J_{n-1} and J_n have opposite signs except for $n = 18, 78, 139, 200, 261, 322, 383$.)

B.3. The triple (4, 2, 3). Here n must be even, $a = d = M = 3$ and $\gamma = 0$. The critical cubic factors, $3W^3 + W^2 - 66W + 112 = (3W - 8)(W^2 + 3W - 14)$, which leads to closed forms: with $\Xi := 4 \log \frac{9+\sqrt{65}}{2} + \frac{1}{2} \log \frac{7+\sqrt{65}}{2}$,

$$r = \Xi - \frac{3}{2} \log 3 = 7.9364908182\dots, \quad s = \Xi - 9 \log 2 + \frac{3}{2} \log 3 = 4.9940030592\dots$$

The relevant critical point lies on the circle $|z| = \sqrt{2}$, and the path consists of two arcs of this circle and two segments through $z = 1$; according to the draft and its audit it was certified numerically, in the audit with ball arithmetic. Further $S \cap [0, 2) = [\frac{1}{4}, \frac{1}{3}) \cup [\frac{3}{2}, \frac{5}{3})$ (period 2) and $\varpi = 0.1257483125\dots$. The bound function gives $6.1000049580\dots$, and already $6.4846910929\dots$ with the multiplier $\text{lcm}(1, \dots, 3n)$, that is without any removable primes. So for this triple even the crudest version of the argument would improve the bound 7.1032 .

B.4. Values of the bound function.

(e_1, e_2, d)	r	s	C	$1 + (r + C)/(s - C)$
(46, 28, 37)	97.4670	60.2778	34.1494	6.037296
(36, 22, 29)	76.3863	47.2173	26.7451	6.037629
(41, 25, 33)	86.9266	53.7475	30.4480	6.037649
(5, 3, 4)	10.5404	6.5301	3.7060	6.044586
(16, 10, 13)	34.2249	21.0953	11.9446	6.045473
(6, 4, 5)	13.1441	8.0310	4.5518	6.086269
(4, 2, 3)	7.9365	4.9940	2.8743	6.100005
(3, 3, 2)	9.1290	4.3878	2.4849	7.103205

In this table s is the value $-\log |f|$ at the relevant critical point. The last column is a proved bound only if a path Γ with $\sup_{\Gamma} |f| = e^{-s}$ has been certified; according to the draft and its audit this was done for (5, 3, 4) and (4, 2, 3) (in the audit with ball arithmetic), and numerically, with a small loss, for (46, 28, 37) and (6, 4, 5). According to numerical scans reported in the draft, the minimum of the bound function over the family is about 6.0373 (the smallest value in the table), so (5, 3, 4) is within 0.008 of the optimum of the family.

B.5. What is special about the configuration. With $q = (z - 1 - i)/(z - 1 + i)$ the end points $1 \pm i$ become $q = 0, \infty$, and all other singular points become 8th roots of unity: $z = \infty, 0 \mapsto q = 1, i$; $z = 1, 2 \mapsto q = -1, -i$; $z = \pm\sqrt{2} \mapsto q = \mp e^{i\pi/4}$; the symmetry becomes $q \mapsto i/q$. This is the reason why 2 is the only prime that needs special treatment.

APPENDIX C. RELATION TO THE INTEGRALS OF ZEILBERGER AND ZUDILIN

Zeilberger and Zudilin [10] use the integrals

$$I_n = i(-1)^{n+1} \int_{-1-2i}^{-1+2i} \frac{5x^{2n}(x^4 + 6x^2 + 25)^{2n}}{(25 - x^2)^{3n+1}} dx,$$

taken along the straight segment (the integrand has poles only at $x = \pm 5$). They are of the same shape as Salikhov's [9]; Bai [2] varies the exponents within this family.

The triple (3, 3, 2) gives the same linear forms. For $n = 1, \dots, 6$ a numerical computation (to more than 50 digits) gives

$$\frac{I_n}{J_n} = \frac{4}{3}, \frac{16}{5}, \frac{160}{21}, \frac{1792}{99}, \frac{6144}{143}, \frac{22528}{221},$$

where J_n is the integral of Appendix B for $(e_1, e_2, d) = (3, 3, 2)$. So the two sequences of linear forms in 1 and π are proportional, with rational factors that grow roughly like $(64/27)^n$. We have no proof of this proportionality, and the two integrands are not related by a Möbius

change of variable: the functions whose n -th powers are integrated, $f_{3,3,2}(z)$ and $x^2(x^4 + 6x^2 + 25)^2/(25 - x^2)^3$, are rational functions of degrees 12 and 10. According to the draft, the same proportionality was observed for other triples with $e_1 = e_2$.

The constants agree. For $(3, 3, 2)$ Claim B.1 has $\gamma = -\frac{1}{2}$, $S = \bigcup_k [k + \frac{1}{3}, k + \frac{1}{2})$, $\varpi = 1.1685237\dots$, $C = 2.4849026\dots$, and

$$r + C = 11.613890045331\dots, \quad s - C = 1.902916485599\dots,$$

$$1 + \frac{r + C}{s - C} = 7.10320533413700172750577342281\dots,$$

which are the constants printed at the end of [10]. The arithmetic of [10] (a different set of removable primes, proved by binomial identities) and that of Claim B.1 thus lead to the same rates for this triple.

An exact link. With $f(W) = (W - 3)^3(W - 2)^3/(W^2 - 8)$ and the function $g(y) = y(y^2 + 6y + 25)^2/(y - 25)^3$ of [10] (where $y = x^2$),

$$f(W) - \frac{27}{64}g(25 - 10W) = \frac{(4W^3 - 5W^2 - 60W + 120)^2(2W^3 - 25W^2 + 96W - 120)}{32W^3(W^2 - 8)}.$$

The squared factor is the critical cubic of f . Hence under $y = 25 - 10W$ the two functions have the same three critical points, with critical values in the ratio $27/64$; correspondingly the cubic $512\lambda^3 - 4719978\lambda^2 + 116694\lambda - 729$ of the critical values of f becomes the cubic $108N^3 - 2359989N^2 + 138304N - 2048$ of [10] under $N = \frac{64}{27}\lambda$. This explains why the two families of forms have the same asymptotic behaviour.

In the family of this paper the exponents e_1 of $(z - 1)(z - 2)$ and e_2 of $z^2 - 2z + 2$ are independent, and the best triples have $e_1 \neq e_2$.

APPENDIX D. A COMPANION: FORMS IN 1, π , $\log 2$

The points $z = 1$ and $z = 2$ are zeros of f that are exchanged by $z \mapsto 2/z$. Let (G, c_n) be the primitive pair of Corollary 4.7. Since $G + c_n \log$ is a primitive of h_n on $\{\operatorname{Re} z > 0\} \setminus \{\sqrt{2}\}$, the integral of h_n along any path from 1 to 2 in this set equals

$$J'_n = u'_n + c_n \log 2, \quad u'_n = G(2) - G(1) \in \mathbb{Q},$$

with the *same* coefficient $c_n = 2v_n$ as in $J_n = u_n + \frac{1}{2}c_n\pi$. According to a companion note to the original draft, $J'_n + iJ_n = 2 \int_1^{1+i} h_n(z) dz$ is a single complex linear form in 1 and $\log(1 + i) = \frac{1}{2}\log 2 + \frac{i\pi}{4}$; the denominators of u'_n are as in Claim B.1 except at the prime 2, where an additional factor $2^{(d-e_2)n/2}$ is needed when $e_2 \leq d$; and Hata's criterion for linear independence [4, Lemma 2.1] would give, for every $\delta > 0$,

$$|a_0 + a_1\pi + a_2 \log 2| \geq H^{-\nu-\delta} \quad \text{for all integers } a_0, a_1, a_2 \text{ with } H = \max(|a_1|, |a_2|) \text{ large,}$$

with $\nu = (r + C')/(s - C')$, $C' = M + \frac{1}{2}(d - e_2)\log 2 - \varpi$: numerically $\nu = 5.8901\dots$ for $(5, 3, 4)$ and $\nu = 5.7596\dots$ for $(6, 4, 5)$, to be compared with $7.016\dots$ in [4, Theorem 1.1]. This is a sketch: the non-vanishing hypothesis of Hata's criterion has only been argued heuristically, and nothing in this appendix has been formalised or audited to the standard of the rest.

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